

Clinic Queue and Appointment Management System

Requirements Specification

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SWE202 - Software Modeling and Design
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1. Executive Summary

1.1 Project Overview

The Clinic Queue and Appointment Management System (CQAMS) is a web-based software solution designed for healthcare clinics that need to manage appointments, walk-in queues, doctor availability, patient records, billing-related information, compliance workflows, and operational reporting from one coordinated platform. The system is intended for clinic administrators, receptionists, doctors, patients, HR officers, finance officers, legal staff, and security staff.

The central business problem addressed by CQAMS is the lack of real-time coordination between appointment scheduling, walk-in queue handling, clinical consultation workflows, and administrative decision-making. By introducing structured role-based access, appointment conflict prevention, queue ticket tracking, automated notifications, audit logging, and reporting, the system aims to reduce waiting time, prevent scheduling conflicts, improve clinic visibility, and protect sensitive patient information.

Team Contribution

The project was completed by the following team members:

Team Leader: Jona Kovaci

Team Members: Ersia Coku, Fatime Fejzo, Klesta Ademaj, Deivid Koci, Geri Gerdoci, Gersi Moli, Skendi Thaci

Jona Kovaci, as the team leader, was responsible for organizing the overall project, dividing the tasks, checking the quality of the work, and preparing the final version for submission. She also reviewed and improved the requirements, diagrams, and documentation to make sure that all parts of the project were clear, consistent, and connected with each other.

The other team members contributed to different parts of the project, including requirements analysis, use cases, activity diagrams, BPMN diagram, ERD, class diagram, interaction diagrams, and documentation support. Each member helped with specific sections, while the team leader collected, corrected, and finalized the work into one complete project.

Overall, the project was developed through teamwork, with Jona Kovaci coordinating the process and ensuring that the final submission followed the required structure and standards.

Team Contribution

Team Leader: Jona Kovaci

Jona Kovaci was responsible for coordinating the whole project, dividing the tasks among the team members, checking the quality of each section, and preparing the final version for submission. She also worked on improving the requirements specification, organizing the diagrams, correcting inconsistencies, and making sure that the final project was complete and well-structured.

Ersia Coku

Ersia Coku contributed to the requirements analysis section. She helped identify the main functional and non-functional requirements of the Clinic Queue and Appointment Management System and supported the organization of the system objectives and scope.

Fatime Fejzo

Fatime Fejzo worked on the use case section. She helped define the main actors of the system, such as Patient, Receptionist, Doctor, and Administrator, and contributed to the description of the main use cases.

Klesta Ademaj

Klesta Ademaj contributed to the Activity Diagram section. She helped describe the main workflow of the system, including patient registration, appointment booking, queue ticket generation, and consultation flow.

Deivid Koci

Deivid Koci worked on the BPMN Diagram section. He helped represent the business process of the clinic system, especially the appointment and queue management process, including decision points and process flow.

Geri Gerdoci

Geri Gerdoci contributed to the ERD section. He helped identify the main database entities, such as Patient, Doctor, Appointment, Queue Ticket, Medical Record, Notification, and Report.

Gersi Moli

Gersi Moli worked on the Class Diagram section. He helped define the main system classes, their attributes, methods, and relationships between users, appointments, queue tickets, medical records, and notifications.

Skendi Thaci

Skendi Thaci contributed to the interaction diagrams and documentation support. He helped with the Sequence Diagram, Collaboration Diagram, and general formatting of the project documentation.

2. Product/Service Description

2.1 Product Context

CQAMS is designed as an independent web-based application following a client-server architecture. Users access the system through a browser-based interface, while the server-side modules coordinate authentication, scheduling, queue management, doctor availability, medical records, notifications, billing, auditing, and reporting. The product is self-contained but may interface with email/SMS notification providers, secure database storage, and optional future integrations such as electronic health record systems or payment gateways.

The product context includes the following major modules: authentication and RBAC, appointment management, real-time queue management, doctor availability, medical record management, HR and finance workflows, legal compliance, security monitoring, reporting, and notification services.

2.2 User Characteristics

User Type	Profile	Experience / Technical Expertise	System Interaction
Administrator	Clinic staff	Basic to intermediate browser skills; role-specific training required.	The Administrator holds the highest level of system authority. This role is responsible for the overall configuration, governance, and maintenance of the system. Core responsibilities include managing all user accounts, assigning and modifying user roles and permissions, configuring system-wide operational settings, generating analytical and operational repo...
Receptionist	Clinic staff	Basic to intermediate browser skills; role-specific training required.	The Receptionist serves as the primary operational interface between patients and the clinic's administrative workflow. This role is responsible for registering new patients, scheduling and modifying appointments, managing the real-time walk-in queue, issuing queue tickets, notifying patients of status changes, and

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			coordinating doctor availability. The Recep...
Doctor	Clinic staff	Basic to intermediate browser skills; role-specific training required.	The Doctor role provides clinical personnel with access to patient-related information and consultation management tools. Doctors are responsible for reviewing their scheduled appointments, accessing patient medical histories and records, updating consultation statuses upon completion, recording clinical notes and diagnoses, prescribing treatments, and analy...
Patient	External clinic user / patient	Basic digital literacy; mobile and browser access expected.	The Patient role enables individuals seeking medical services to engage with the clinic's scheduling and information services through a self-service interface. Patients may register personal accounts, book or cancel appointments, view their real-time queue position and estimated waiting time, access their personal medical records and appointment history, and...
HR Officer	Clinic staff	Basic to intermediate browser skills; role-specific training required.	The HR Officer is responsible for managing staff-related information within the system. This role oversees the onboarding and offboarding of clinic personnel, maintains employee records, manages work schedules and leave requests, and coordinates with the Administrator regarding user account provisioning. The HR Officer interacts with the staff management mod...
Finance Officer	Clinic staff	Basic to intermediate browser skills; role-specific training required.	The Finance Officer is responsible for managing the financial operations of the clinic as reflected within the system. This role oversees billing records, processes patient invoices, monitors payment statuses, generates financial reports, and liaises with the Administrator for budget-related configuration. The Finance Officer has access to appointment data f...
Legal Staff	Clinic staff	Basic to intermediate browser skills; role-specific training required.	The Legal Staff role is responsible for ensuring that the clinic's use of the system complies with applicable laws, regulations, and institutional policies. This role reviews audit logs and data-access records for compliance purposes, manages consent documentation, and is notified of any data-protection incidents or policy violations flagged by the system. L...
Security Staff	Clinic staff	Basic to intermediate browser skills; role-specific training required.	The Security Staff role is responsible for monitoring system access, detecting anomalous activity, and responding to potential security incidents within the platform. This role has access to the security monitoring dashboard, login audit logs, and access-control reports. Security Staff may escalate incidents to the Administrator and coordinate with Legal Sta...

2.3 Assumptions

- All users possess a basic level of digital literacy sufficient to operate a standard web-based interface.
- The clinic operates within a single physical location or a limited number of satellite branches.
- Stable internet connectivity is available at all points of deployment.
- The system will comply with applicable healthcare data-protection regulations, including the General Data Protection Regulation (GDPR) where applicable.
- Minors under the age of 18 are represented by a guardian account; the system does not permit independent account creation for minors.
- All financial data processed by the system is subject to applicable fiscal and healthcare billing regulations.

2.4 Constraints and Dependencies

- **Web-based deployment:** The product requires stable internet connectivity and modern browser support at clinic workstations and patient devices.
- **Data protection:** Sensitive patient and clinical data must be processed according to GDPR and applicable healthcare privacy regulations.
- **Role-based security:** Access management, audit logs, and secure authentication must be completed before clinical records and administrative modules are released.
- **Operational criticality:** Downtime directly affects appointment scheduling, queue control, and patient communication; therefore maintenance windows must be communicated in advance.
- **Module dependency:** Doctor availability and master schedule data must exist before appointment booking and queue allocation can operate correctly.
- **External services:** Automated reminders and queue alerts depend on reliable notification delivery through email/SMS or equivalent messaging services.

3. Requirements

All requirements are numbered and prioritised using the template priority scheme: Priority 1 = must-have / policy or legal requirement; Priority 2 = needed for improved processing and immediate operational benefit; Priority 3 = nice-to-have or extended functionality.

3.1 Functional Requirements

Req#	Requirement	Comments	Priority	Date Rvw'd	SME Reviewed / Approved
FR-01	Administrator creates new user accounts and assigns appropriate system roles to clinic staff.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-02	Administrator deactivates user accounts when a staff member's employment or role is terminated.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-03	Administrator configures system-wide authentication policies, including password complexity rules and session timeout durations.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-04	Administrator reviews the system audit log to monitor user login events and detect unauthorized access attempts.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-05	Receptionist logs into the system using unique credentials through the secure authentication portal.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-06	Doctor logs into the system and accesses only the modules and data assigned to the Doctor role.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-07	Patient registers a personal account by providing the required demographic and contact information.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-08	Patient logs into the patient portal using a registered email address and a verified password.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-09	Security Staff monitors all user login events and active sessions through the security dashboard to detect anomalous access patterns.	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-10	Legal Staff reviews data-access audit logs to verify compliance with applicable data-protection	Business Process = Authentication and Access Control	1	14/03/2026	Jona Kovaci / CQAMS Team

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	regulations and institutional policies.				
FR-11	Receptionist schedules a new patient appointment by selecting a doctor, date, and available time slot.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-12	Receptionist modifies an existing appointment upon patient request, subject to current doctor availability.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-13	Receptionist cancels a scheduled appointment and updates the queue and the doctor's calendar accordingly.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-14	Receptionist views a consolidated daily appointment schedule for all doctors registered in the system.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-15	Patient books a new appointment through the self-service portal by selecting a preferred doctor and time slot.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-16	Patient cancels a previously booked appointment within the permitted cancellation window.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-17	Patient views a list of upcoming and past appointments in the personal account dashboard.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-18	Doctor reviews the daily appointment schedule upon logging into the system.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-19	Doctor requests the rescheduling of an appointment in the event of an unavoidable scheduling conflict.	Business Process = Appointment Scheduling	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-20	Receptionist registers a walk-in patient into the real-time queue and issues an automatically generated queue ticket.	Business Process = Queue Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-21	Receptionist removes a patient from the active queue upon departure, no-show, or voluntary withdrawal.	Business Process = Queue Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-22	Receptionist assigns an urgent priority flag to a patient in the queue when the clinical situation warrants expedited service.	Business Process = Queue Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-23	Patient views their current queue position and estimated waiting time through the patient portal.	Business Process = Queue Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-24	Doctor marks a consultation as complete, which triggers automatic advancement of the queue to the next patient.	Business Process = Queue Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-25	Doctor views the real-time queue dashboard for their assigned consultation room upon logging in.	Business Process = Queue Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-26	Doctor analyzes urgent cases flagged in the queue and determines the appropriate priority level and immediate clinical action required.	Business Process = Queue Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-27	Doctor defines and manages weekly availability by specifying working hours, available days, and marking specific dates as unavailable due to leave, professional	Business Process = Doctor Schedule and Availability Management	2	14/03/2026	Jona Kovaci / CQAMS Team

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	development, or other commitments.				
FR-28	Administrator assigns doctors to specific clinic departments or medical service categories within the system.	Business Process = Doctor Schedule and Availability Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-29	Receptionist checks doctor availability in real time before scheduling or modifying a patient appointment.	Business Process = Doctor Schedule and Availability Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-30	Administrator manages the clinic's master schedule, including operating hours, public holidays, and shift configurations.	Business Process = Doctor Schedule and Availability Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-31	HR Officer manages staff leave requests and updates doctor and clinical staff availability records within the scheduling module.	Business Process = Doctor Schedule and Availability Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-32	Receptionist creates a new patient profile containing verified demographic, contact, and health insurance information.	Business Process = Patient Records Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-33	Doctor accesses the complete medical history and previous consultation records of a scheduled patient, as specified in Use Case UC-07 (View Patient Record).	Business Process = Patient Records Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-34	Doctor records consultation notes, clinical diagnoses, and prescribed treatments in the patient's medical record.	Business Process = Patient Records Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-35	Doctor updates the patient's medical record upon completion of each consultation session.	Business Process = Patient Records Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-36	Patient views personal health records, past diagnoses, prescriptions, and consultation summaries through the patient portal.	Business Process = Patient Records Management	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-37	Legal Staff manages patient consent documentation and ensures that data collection and processing activities comply with applicable legal requirements.	Business Process = Patient Records Management	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-38	HR Officer registers new clinic staff members in the system and initiates the account provisioning process with the Administrator.	Business Process = Human Resources and Finance	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-39	HR Officer maintains employee records including personal data, employment contracts, qualifications, and performance history.	Business Process = Human Resources and Finance	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-40	HR Officer generates HR reports on staff attendance, leave balances, and scheduling compliance for management review.	Business Process = Human Resources and Finance	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-41	Finance Officer generates patient invoices based on completed consultations and services rendered as recorded in the appointment system.	Business Process = Human Resources and Finance	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-42	Finance Officer monitors outstanding payment statuses and follows up on unpaid invoices	Business Process = Human Resources and Finance	2	14/03/2026	Jona Kovaci / CQAMS Team

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	through the billing module.				
FR-43	Finance Officer generates monthly and annual financial reports covering clinic revenue, billing volumes, and payment collection rates.	Business Process = Human Resources and Finance	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-44	System sends an automated appointment confirmation notification to the patient immediately following a booking.	Business Process = Notifications	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-45	System sends an automated appointment reminder to the patient 24 hours prior to the scheduled appointment time.	Business Process = Notifications	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-46	System notifies the patient when their queue position advances to within the next two consultations.	Business Process = Notifications	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-47	Receptionist sends a manual notification to a patient regarding appointment changes, cancellations, or urgent instructions.	Business Process = Notifications	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-48	System notifies the Doctor when a new appointment is added to or removed from their daily calendar.	Business Process = Notifications	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-49	System alerts Security Staff and the Administrator when a suspicious login attempt or unauthorized access event is detected.	Business Process = Notifications	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-50	Administrator generates daily and monthly operational reports covering appointment volumes, cancellation rates, and no-show statistics.	Business Process = Reporting and System Administration	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-51	Administrator exports operational and analytical reports in standardised formats, including PDF and CSV, for institutional review.	Business Process = Reporting and System Administration	3	14/03/2026	Jona Kovaci / CQAMS Team
FR-52	Administrator monitors system performance metrics and usage statistics through a dedicated administrative dashboard.	Business Process = Reporting and System Administration	3	14/03/2026	Jona Kovaci / CQAMS Team
FR-53	Administrator manages system configuration parameters including service categories, clinic locations, and notification templates.	Business Process = Reporting and System Administration	2	14/03/2026	Jona Kovaci / CQAMS Team
FR-54	System maintains a complete and tamper-evident audit trail of all modifications made to patient records and appointment data.	Business Process = Reporting and System Administration	1	14/03/2026	Jona Kovaci / CQAMS Team
FR-55	Security Staff escalates detected security incidents to the Administrator and Legal Staff for investigation and resolution in accordance with the clinic's incident response policy.	Business Process = Reporting and System Administration	1	14/03/2026	Jona Kovaci / CQAMS Team

3.2 Non-Functional Requirements

The non-functional requirements define the system quality attributes and operational constraints. They are grouped by the categories requested in the template and by the categories already documented in the CQAMS requirements work.

3.2.1 Product Requirements

Req#	Category	Requirement	Priority	Verification Criteria
NFR-01	Security	The system shall enforce role-based access control (RBAC), ensuring that each user can access only the modules and data corresponding to their assigned role.	1	Test / audit / inspection based on measurable threshold.
NFR-02	Security	The system shall encrypt all data transmitted between the client and the server using TLS 1.2 or higher.	1	Test / audit / inspection based on measurable threshold.
NFR-03	Security	The system shall store all user passwords using a recognised cryptographic hashing algorithm such as bcrypt or Argon2; plaintext password storage is strictly prohibited.	1	Test / audit / inspection based on measurable threshold.
NFR-04	Security	The system shall automatically invalidate and terminate an authenticated user session after 15 minutes of inactivity.	1	Test / audit / inspection based on measurable threshold.
NFR-05	Security	The system shall record and retain a comprehensive audit log of all user authentication events, data modifications, and administrative actions for a minimum of 12 months.	1	Test / audit / inspection based on measurable threshold.
NFR-06	Security	The system shall restrict access to sensitive patient records so that only authorised roles — Doctor, Administrator, and Legal Staff — may view clinical data, in compliance with applicable data-protection policies and the principle of least privilege.	1	Test / audit / inspection based on measurable threshold.
NFR-07	Security	The system shall provide Security Staff with a real-time security monitoring dashboard displaying active sessions, failed login attempts, and flagged access anomalies.	1	Test / audit / inspection based on measurable threshold.
NFR-08	Performance	The system shall process and confirm an appointment booking request within two seconds under standard load conditions.	2	Test / audit / inspection based on measurable threshold.
NFR-09	Performance	The system shall render any dashboard view or patient record page within three seconds under typical network conditions.	2	Test / audit / inspection based on measurable threshold.
NFR-10	Performance	The system shall support a minimum of 100 concurrent authenticated users without measurable degradation in response time.	2	Test / audit / inspection based on measurable threshold.
NFR-11	Performance	The real-time queue dashboard shall refresh at intervals not exceeding ten seconds to reflect the current queue status.	2	Test / audit / inspection based on measurable threshold.
NFR-12	Performance	Database queries critical to daily operations shall return results within one second for datasets of up to 10,000 records.	2	Test / audit / inspection based on measurable threshold.
NFR-13	Usability	The system interface shall conform to established web usability standards, enabling a newly trained Receptionist to complete core appointment tasks after no more than one hour of guided orientation.	2	Test / audit / inspection based on measurable threshold.
NFR-14	Usability	The patient portal shall be fully responsive and accessible on mobile devices, tablets, and desktop browsers without loss of functionality.	2	Test / audit / inspection based on measurable threshold.
NFR-15	Usability	The system shall provide real-time inline validation and descriptive error messages to assist users in correcting form input errors prior to submission.	2	Test / audit / inspection based on measurable threshold.
NFR-16	Usability	The system shall support a minimum of two interface languages — English and Albanian — to reflect the linguistic profile of the target user base.	2	Test / audit / inspection based on measurable threshold.

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NFR-17	Usability	All irreversible actions shall be preceded by an on-screen confirmation dialog requiring explicit user acknowledgment before execution.	2	Test / audit / inspection based on measurable threshold.
NFR-18	Reliability	The system shall achieve a minimum monthly uptime of 99.5%, excluding pre-scheduled and communicated maintenance windows.	2	Test / audit / inspection based on measurable threshold.
NFR-19	Reliability	The system shall perform automated incremental data backups at daily intervals, with a minimum retention period of 90 days.	2	Test / audit / inspection based on measurable threshold.
NFR-20	Reliability	In the event of an unplanned system failure, the system shall be fully recoverable from the most recent backup within four hours.	2	Test / audit / inspection based on measurable threshold.
NFR-21	Reliability	The system shall implement transactional integrity controls to prevent partial updates, data corruption, or inconsistent states arising from concurrent operations.	2	Test / audit / inspection based on measurable threshold.
NFR-22	Scalability	The system architecture shall support horizontal scaling to accommodate growth in registered patients, clinical staff, and concurrent users without requiring architectural redesign.	2	Test / audit / inspection based on measurable threshold.
NFR-23	Scalability	The database layer shall be capable of storing a minimum of 500,000 patient records and 1,000,000 appointment entries without performance degradation.	2	Test / audit / inspection based on measurable threshold.
NFR-24	Scalability	The system shall be extensible to incorporate additional clinic branches, departments, or service categories through configuration, with minimal development effort required.	2	Test / audit / inspection based on measurable threshold.
NFR-25	Availability	The system shall be available 24 hours per day, seven days per week, with the exception of pre-announced maintenance periods not exceeding four hours per calendar month.	1	Test / audit / inspection based on measurable threshold.
NFR-26	Availability	All planned maintenance activities shall be scheduled outside peak clinic operating hours and communicated to administrative users a minimum of 48 hours in advance.	1	Test / audit / inspection based on measurable threshold.
NFR-27	Maintainability	The system codebase shall adhere to documented coding standards and include sufficient inline documentation to enable an unfamiliar developer to undertake maintenance tasks within a reasonable onboarding period.	2	Test / audit / inspection based on measurable threshold.
NFR-28	Maintainability	The system shall be constructed using a modular architecture, allowing individual components to be updated, tested, or replaced in isolation without disrupting the remainder of the application.	2	Test / audit / inspection based on measurable threshold.
NFR-29	Maintainability	The system shall produce structured application logs — including error events, warnings, and key operational milestones — accessible to system administrators for diagnostic and proactive maintenance purposes.	2	Test / audit / inspection based on measurable threshold.
NFR-30	Legal and Regulatory Compliance	The system shall comply with the General Data Protection Regulation (GDPR) and all applicable national healthcare data-protection legislation governing the storage, processing, and transmission of patient data.	1	Test / audit / inspection based on measurable threshold.
NFR-31	Legal and Regulatory Compliance	The system shall provide Legal Staff with the ability to generate compliance reports and extract	1	Test / audit / inspection based on measurable threshold.

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		consent records and data-access histories upon request.		
NFR-32	Legal and Regulatory Compliance	The system shall enforce documented data retention and deletion policies, ensuring that patient records are retained for the legally required minimum period and are securely deleted upon expiry of the retention window.	1	Test / audit / inspection based on measurable threshold.
NFR-33	Legal and Regulatory Compliance	The system shall notify Legal Staff and the Administrator automatically in the event of a suspected data-protection incident or unauthorised disclosure of personal data.	1	Test / audit / inspection based on measurable threshold.

3.2.1.1 Usability Requirements

The interface shall be easy to learn, responsive across desktop and mobile screens, bilingual in English and Albanian, and supported by clear inline validation and confirmation prompts for irreversible actions.

3.2.1.2 Performance Requirements

Appointment booking shall be processed within two seconds under standard load, dashboard and patient record pages shall render within three seconds, and queue status shall refresh at intervals not exceeding ten seconds.

3.2.1.3 Availability

The system shall provide 24/7 access with at least 99.5% monthly uptime, excluding scheduled maintenance not exceeding four hours per calendar month.

3.2.1.4 Security

The system shall enforce RBAC, TLS 1.2+ transport encryption, secure password hashing, inactivity timeout after 15 minutes, audit log retention for at least 12 months, and least-privilege access to patient records.

3.2.2 Organizational Requirements

CQAMS shall follow the project team's GitHub-based version control process and academic documentation workflow. Requirements, use cases, diagrams, ERD, state diagrams, sequence/collaboration diagrams, and design pattern deliverables are maintained as separate artefacts and consolidated in this final specification.

3.2.3 External Requirements

The system shall comply with GDPR and applicable national healthcare data-protection rules. It shall also be designed to support external notification services and future integrations with electronic health records, payment services, or reporting exports without major architectural redesign.

4. User Scenarios/Use Cases

The following use cases summarise the major user-facing and system-facing functions of CQAMS. The naming follows the template convention where functional requirements are mapped to UC identifiers and then to diagrams in the diagram section.

Use Case	Mapped Requirement(s)	Primary / Secondary Actors	Summary	Preconditions / Dependency
UC-01: Manage User Accounts	FR-01, FR-02	Primary Actor: Administrator.	The Administrator creates, activates, and deactivates user accounts for all clinic staff and assigns each account the appropriate system role.	The Administrator is authenticated and has system-configuration privileges.
UC-02: Configure Authentication Policies	FR-03	Primary Actor: Administrator.	The Administrator configures system-wide authentication settings, including password complexity rules and session timeout durations.	The Administrator is authenticated and has system-configuration privileges.
UC-03: Review System Audit Log	FR-04, FR-10, FR-54	Primary Actor: Administrator. Secondary Actor: Legal Staff.	The Administrator and Legal Staff review the system audit log to monitor user login events, data-access records, and administrative actions for compliance and security purposes.	The Actor is authenticated. Audit log entries exist in the system.

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UC-04: User Login	FR-05, FR-06, FR-08	Primary Actor: Any authenticated staff user.	All staff users (Receptionist, Doctor, HR Officer, Finance Officer, Legal Staff, Security Staff) authenticate themselves using their unique credentials and access their role-specific dashboard.	The user has an active account in the system. The system is accessible.
UC-05: Patient Registration and Login	FR-07, FR-08	Primary Actor: Patient.	A patient registers a new personal account and subsequently logs into the patient portal to access self-service features.	For registration: no existing account associated with the provided email address. For login: an active patient account exists.
UC-06: Security Staff Monitors Login Events	FR-09, FR-49	Primary Actor: Security Staff.	Security Staff monitor all user login events and active sessions through the security dashboard to detect anomalous access patterns and respond to potential security incidents.	Security Staff member is authenticated.
UC-07: Schedule and Modify Appointment	FR-11, FR-12, FR-29	Primary Actor: Receptionist. Secondary Actor: Patient (notified).	The Receptionist schedules a new patient appointment or modifies an existing one, selecting the appropriate doctor, date, and available time slot.	The Receptionist is authenticated. A patient profile exists in the system. The selected doctor has at least one available time slot.
UC-08: Cancel Appointment	FR-13	Primary Actor: Receptionist. Secondary Actors: Patient (notified), Doctor (notified).	The Receptionist cancels a scheduled appointment at the request of the patient or due to operational necessity, updating the queue and the doctor's calendar accordingly.	The Receptionist is authenticated. The appointment exists and has not yet taken place.
UC-09: Patient Books or Cancels Appointment	FR-15, FR-16	Primary Actor: Patient.	A patient books a new appointment or cancels an existing one through the self-service patient portal.	The patient is authenticated in the patient portal.
UC-10: View Appointment Schedule and History	FR-14, FR-17, FR-18	Primary Actor: Receptionist, Doctor, or Patient (depending on context).	The Receptionist views a consolidated daily schedule for all doctors; the Doctor reviews their own daily schedule upon login; the Patient views their list of upcoming and past appointments.	The Actor is authenticated.
UC-11: Request Appointment Rescheduling	FR-19	Primary Actor: Doctor. Secondary Actor: Receptionist.	A Doctor requests the rescheduling of an appointment due to an unavoidable conflict. The request is routed to the Receptionist for action.	The Doctor is authenticated. The appointment exists and has not yet occurred.
UC-12: Register Walk-in Patient to Queue	FR-20	Primary Actor: Receptionist. Secondary Actor: Patient.	The Receptionist registers a walk-in patient into the real-time queue and issues a unique queue ticket.	The Receptionist is authenticated. The queue module is active.
UC-13: Remove Patient from Queue	FR-21	Primary Actor: Receptionist.	The Receptionist removes a patient from the active queue following departure, a no-show, or voluntary withdrawal.	The Receptionist is authenticated. The patient is currently present in the active queue.
UC-14: Assign Urgent Priority Flag	FR-22	Primary Actor: Receptionist. Secondary Actor: Doctor (notified).	The Receptionist assigns an urgent priority flag to a patient in the queue when the clinical situation warrants immediate attention.	The Receptionist is authenticated. The patient is present in the active queue.
UC-15: Patient Views Queue Position	FR-23	Primary Actor: Patient.	A patient views their current real-time queue position and estimated waiting time through the patient portal.	The patient is authenticated and currently registered in the active clinic queue.
UC-16: Doctor Completes Consultation and Analyzes Urgent Cases	FR-24, FR-25, FR-26	Primary Actor: Doctor. Secondary Actor: Receptionist.	The Doctor marks a completed consultation as done, triggering automatic queue advancement to the next patient. The Doctor also reviews and analyzes any urgent cases flagged in the queue to determine the appropriate clinical response.	The Doctor is authenticated. At least one patient is active in the consultation queue.
UC-17: Doctor Defines and Manages Availability	FR-27	Primary Actor: Doctor. Secondary Actor: Administrator.	The Doctor defines their weekly working hours, available days, and marks specific dates as unavailable due to leave, professional development, or other commitments.	The Doctor is authenticated.
UC-18: Assign Doctor to Department and Check Availability	FR-28, FR-29	Primary Actor: Administrator (assignment); Receptionist (availability check).	The Administrator assigns a doctor to a specific clinic department or medical service category. The Receptionist checks real-time doctor availability before scheduling or modifying an appointment.	The Administrator is authenticated and the Doctor's account exists (assignment). The Receptionist is authenticated (availability check).
UC-19: Manage Clinic Master Schedule	FR-30, FR-31	Primary Actor: Administrator (master schedule); HR Officer (leave management).	The Administrator manages the clinic's master schedule, configuring operating hours, public holidays, and shift patterns. The HR Officer manages staff leave requests and updates availability records.	The Actor is authenticated and has the appropriate role.
UC-20: Create and Manage Patient Profile	FR-32	Primary Actor: Receptionist.	The Receptionist creates a new patient profile containing verified demographic, contact, and health insurance information.	The Receptionist is authenticated. The patient does not already have a profile in the system.
UC-21: Access and Update Patient Medical Records	FR-33, FR-34, FR-35, FR-36	Primary Actor: Doctor. Secondary Actors: Patient (read access).	The Doctor accesses the complete medical history and consultation records of a scheduled patient (as specified in Use Case UC-21 – Access Patient Medical History), and records consultation notes,	The Doctor is authenticated. The patient is scheduled for consultation. Patient consent is recorded in the system.

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			diagnoses, and prescribed treatments following each consultation. The Patient views their personal records through the portal.	
UC-22: Manage Patient Consent Documentation	FR-37	Primary Actor: Legal Staff. Secondary Actor: Receptionist.	Legal Staff manage patient consent documentation, ensuring all data collection and processing activities comply with applicable legal requirements.	The Legal Staff member is authenticated.
UC-23: HR Officer Manages Staff Records and Leave	FR-38, FR-39, FR-40	Primary Actor: HR Officer. Secondary Actor: Administrator.	The HR Officer registers new clinic staff, maintains employee records including personal data and qualifications, manages leave requests, and generates HR reports.	The HR Officer is authenticated.
UC-24: Finance Officer Manages Billing and Financial Reports	FR-41, FR-42, FR-43	Primary Actor: Finance Officer.	The Finance Officer generates patient invoices based on completed consultations, monitors outstanding payment statuses, and generates monthly and annual financial reports.	The Finance Officer is authenticated.
UC-25: Automated System Notifications	FR-44, FR-45, FR-46, FR-48	Primary Actor: System (automated). Secondary Actors: Patient, Doctor.	The system automatically sends appointment confirmation notifications, 24-hour reminders, queue advancement alerts, and Doctor calendar change notifications without requiring manual intervention.	A triggering event has occurred: appointment creation, appointment time approaching, queue position change, or calendar modification.
UC-26: Receptionist Sends Manual Notification	FR-47	Primary Actor: Receptionist. Secondary Actor: Patient.	The Receptionist sends a manual notification to a patient regarding appointment changes, cancellations, or urgent instructions.	The Receptionist is authenticated.
UC-27: Generate and Export Operational Reports	FR-50, FR-51	Primary Actor: Administrator.	The Administrator generates daily and monthly operational reports covering appointment volumes, cancellation rates, and no-show statistics, and exports them in standardised formats.	The Administrator is authenticated.
UC-28: Monitor System Performance	FR-52	Primary Actor: Administrator.	The Administrator monitors system performance metrics and usage statistics through the administrative dashboard to identify and address performance issues.	The Administrator is authenticated.
UC-29: Security Staff Escalates Incident	FR-55	Primary Actor: Security Staff. Secondary Actors: Administrator, Legal Staff.	Security Staff detect a security incident and escalate it to the Administrator and Legal Staff for investigation and resolution.	Security Staff is authenticated. A security incident has been identified in the system.

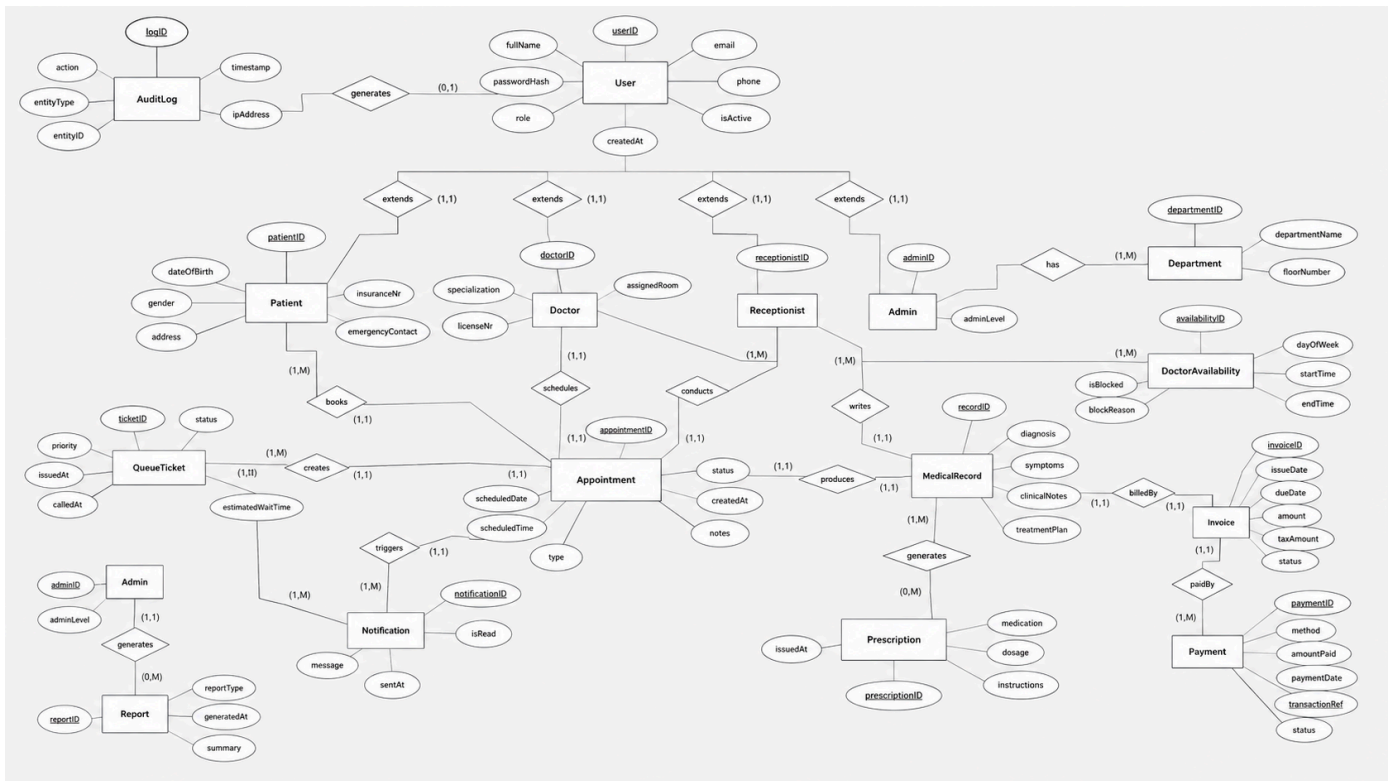
5. Diagrams

This section consolidates the diagrams prepared throughout the course. The diagram identifiers follow the relationship requested by the template: requirements map to use cases, activity diagrams, sequence diagrams, and collaboration diagrams where applicable.

5.1 ER Diagram

The ER diagram models the system data layer around User, Patient, Doctor, Appointment, Queue Ticket, Medical Record, Prescription, Notification, Audit Log, and Doctor Availability entities. It supports RBAC, appointment history, queue tracking, medical documentation, reminders, and auditability.

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5.2 Use Case Diagram (general)

The general use case diagram includes 8 actors and 29 use cases across authentication, appointment scheduling, queue management, doctor availability, patient records, HR/finance, notifications, reporting, and security escalation.

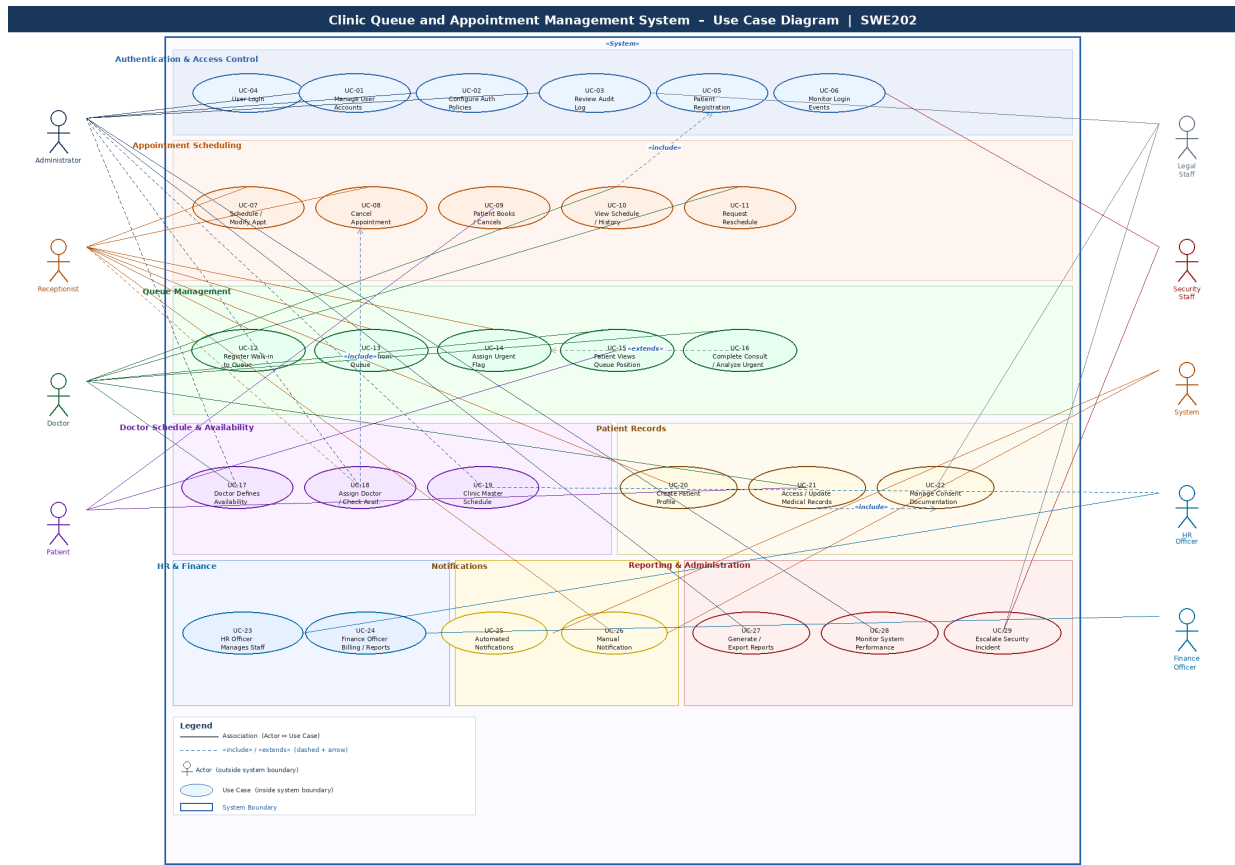
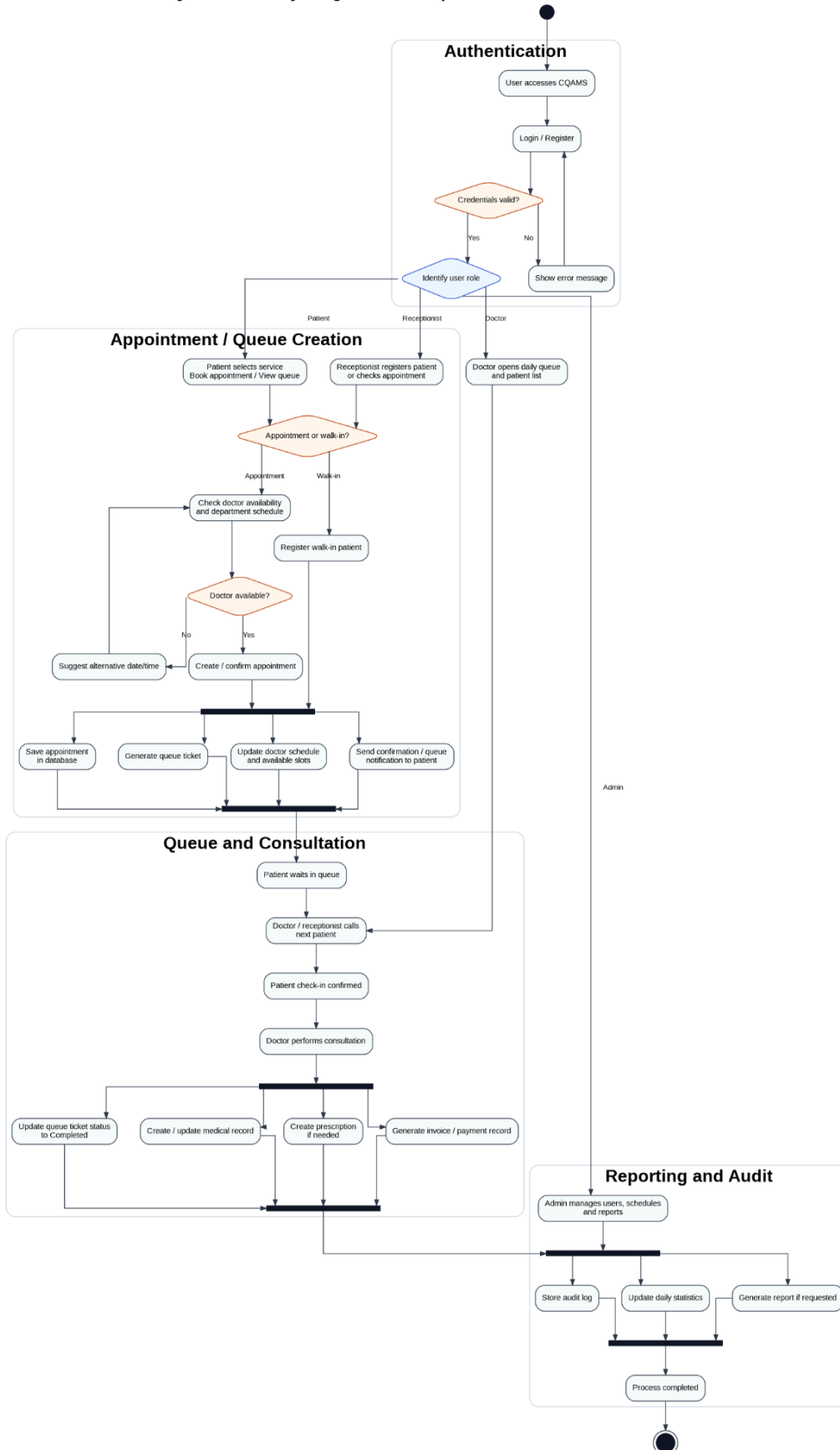


Figure 5.2 - Complete UML Use Case Diagram

5.3 Activity Diagram

The activity/BPMN diagrams document the main process flows for appointment booking, walk-in queue management, and consultation/medical record update. They show the interaction between patient, receptionist, doctor, system modules, availability checks, and notification service.

CQAMS - Activity Diagram with Dependent / Fork and Join Nodes



CQAMS Requirements Specification

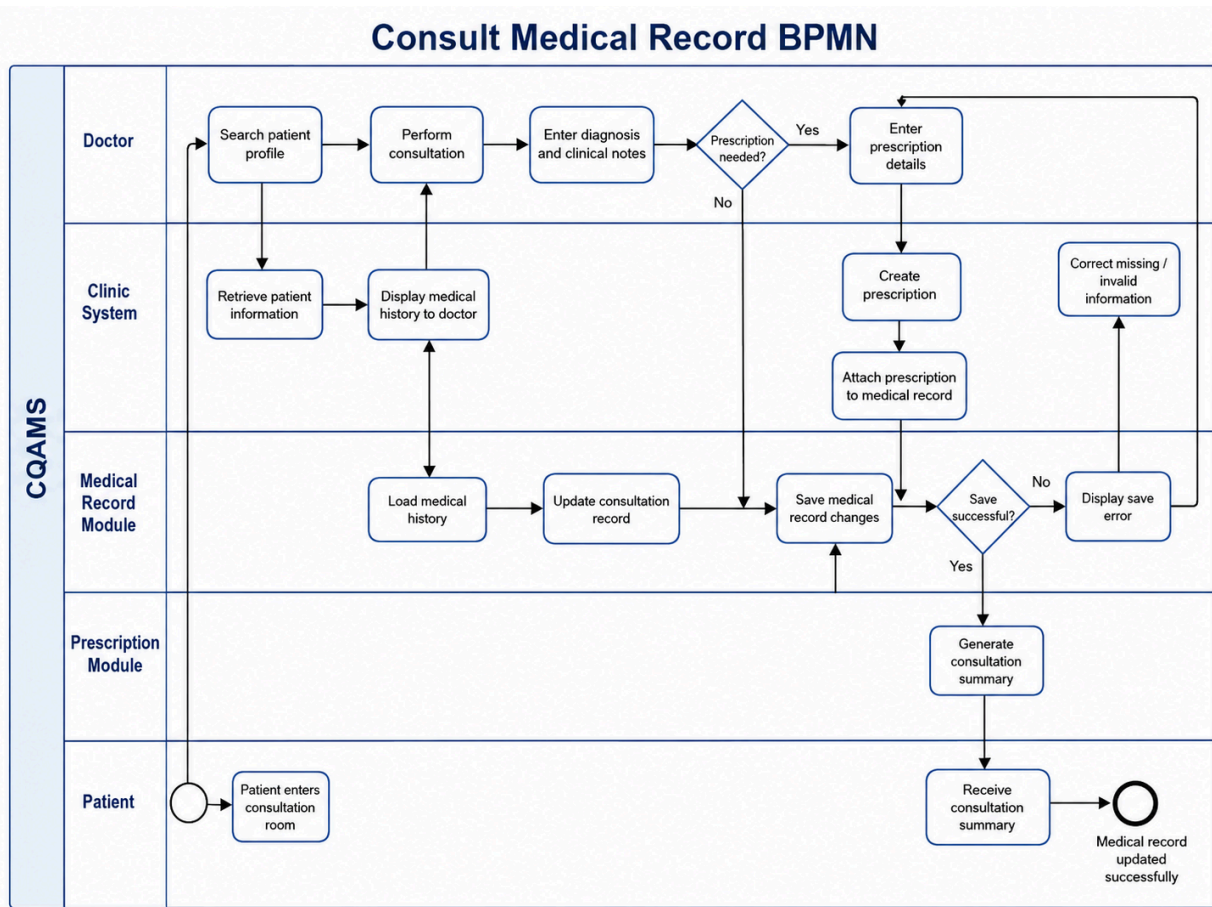
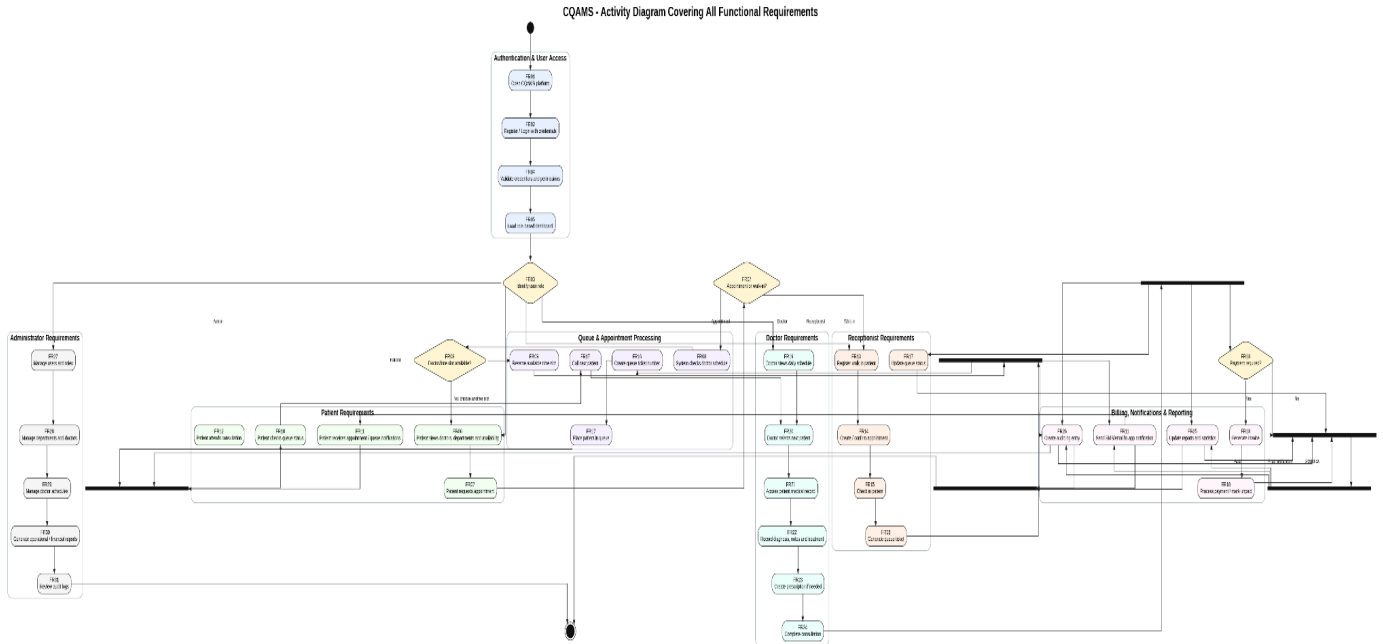
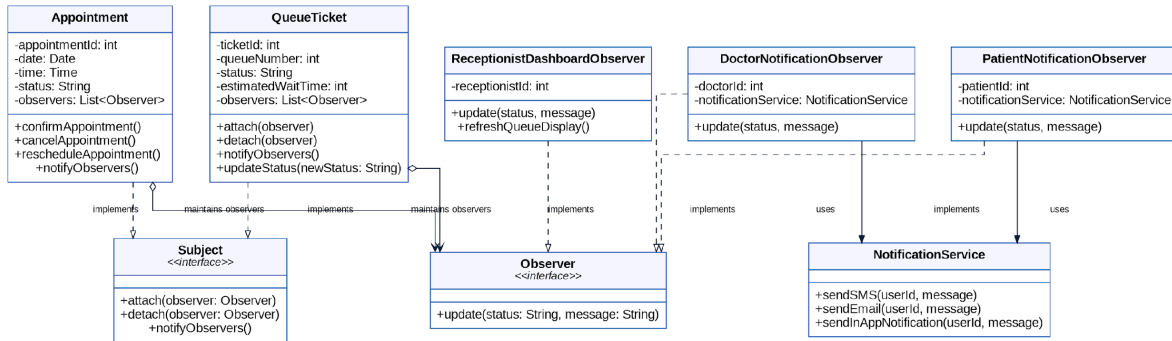


Figure 5.3.3 - AC-21 Consultation and Medical Record BPMN / Activity Flow

5.4 Class diagram.

The general class structure is represented through the project class diagrams and pattern-specific refinements. Core classes include User, Patient, Doctor, Receptionist, Administrator, Appointment, QueueTicket, MedicalRecord, Prescription, NotificationService, QueueManagementSystem, and supporting service classes.



Observer Pattern - CQAMS Status Notification Model

CQAMS Requirements Specification

Clinic Queue and Appointment Management System (CQAMS) - Detailed UML Class Diagram

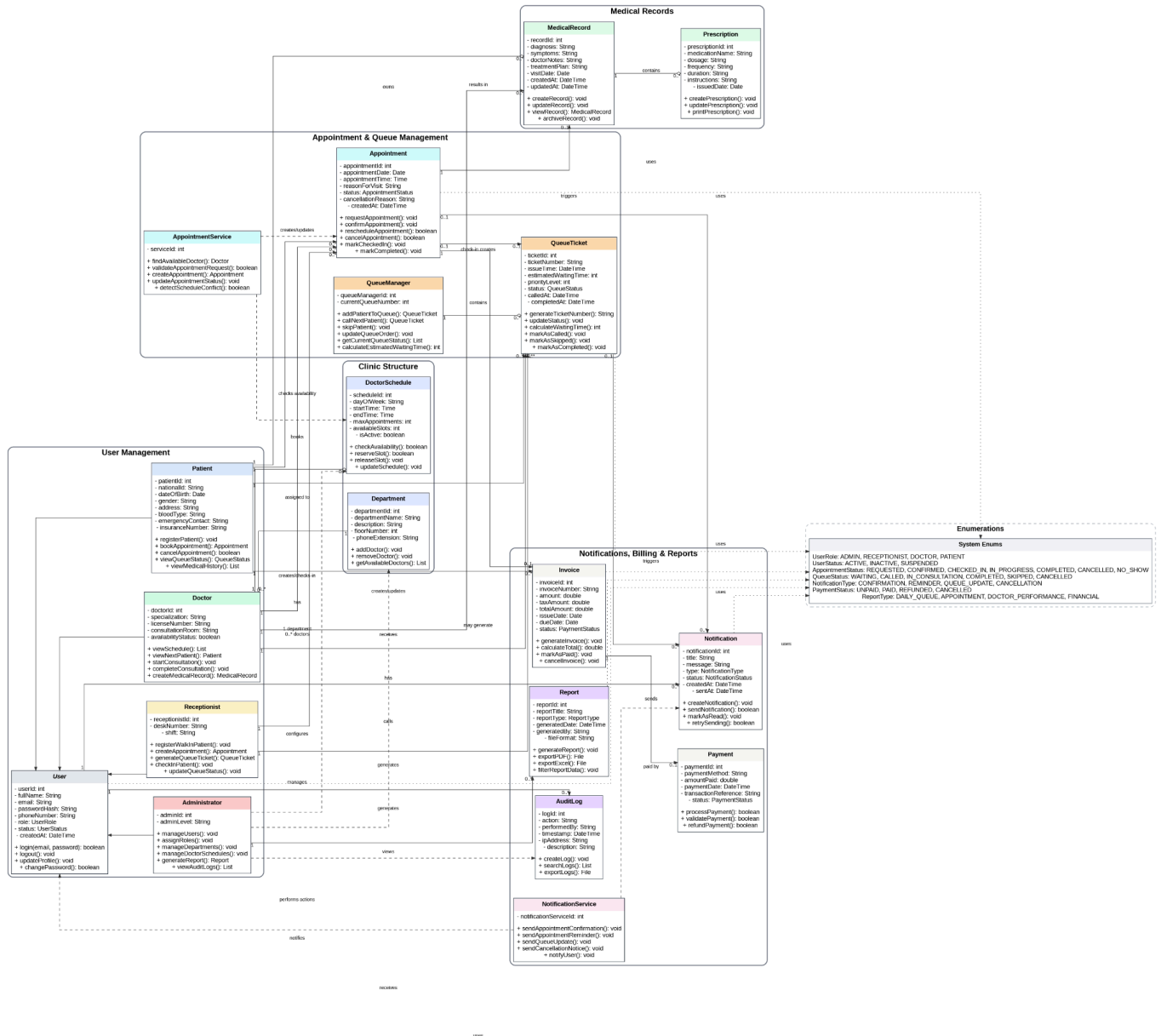


Figure 5.4 - Class Diagram excerpt from project design pattern work

5.5 State diagram

The state diagrams describe lifecycle behaviour for Patient, Doctor, Receptionist, and Administrator roles. These diagrams help verify valid state transitions and prevent invalid workflow states during use of the application.

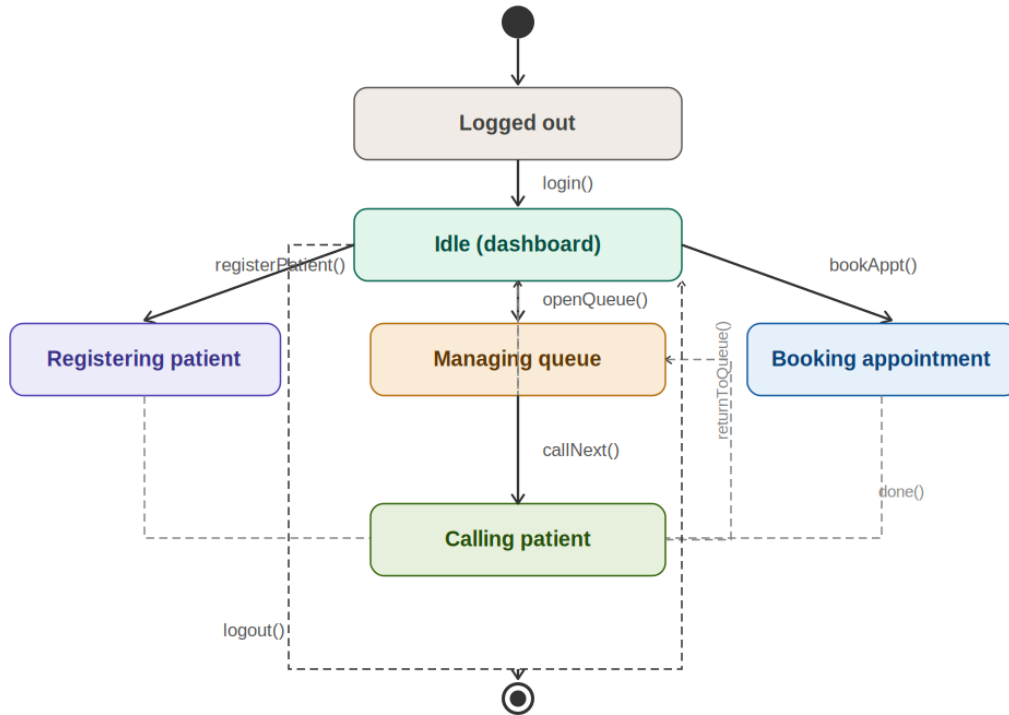
Receptionist — State Diagram

Figure 5.5.1 - Patient State Diagram

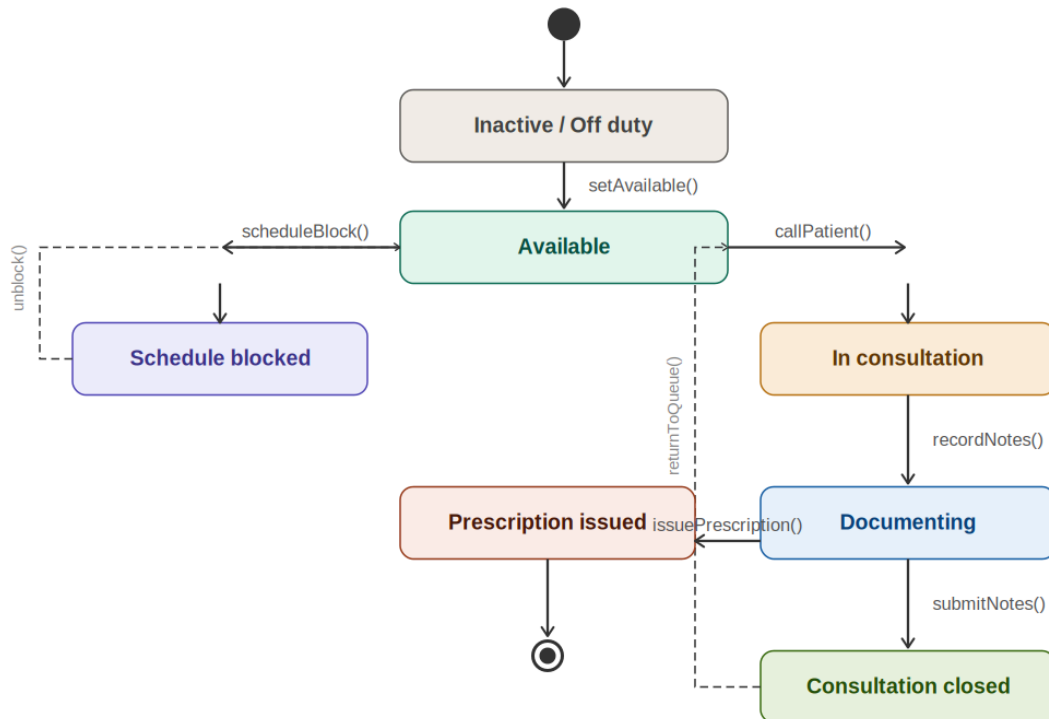
Doctor — State Diagram

Figure 5.5.2 - Doctor State Diagram

Administrator — State Diagram

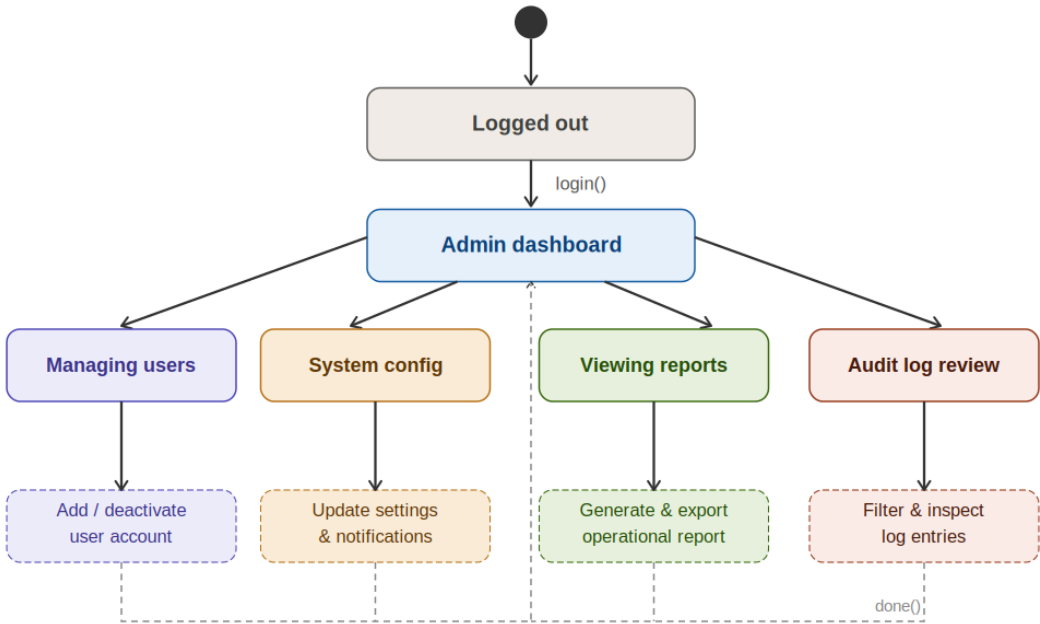


Figure 5.5.3 - Receptionist State Diagram

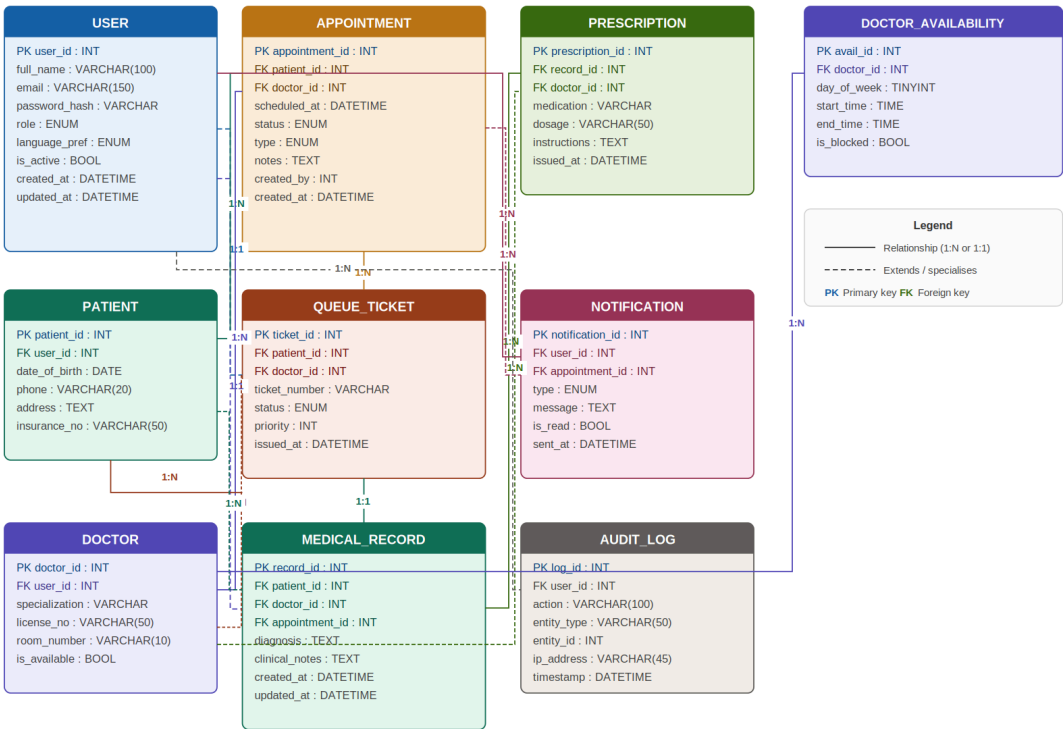


Figure 5.5.4 - Administrator State Diagram

5.6 Sequence diagram.

The sequence diagram below represents queue management interactions among Receptionist, Patient, Doctor, QueueController, Ticket, and NotificationService. It follows the AC-12 queue activity flow and is therefore referenced as SC-12.

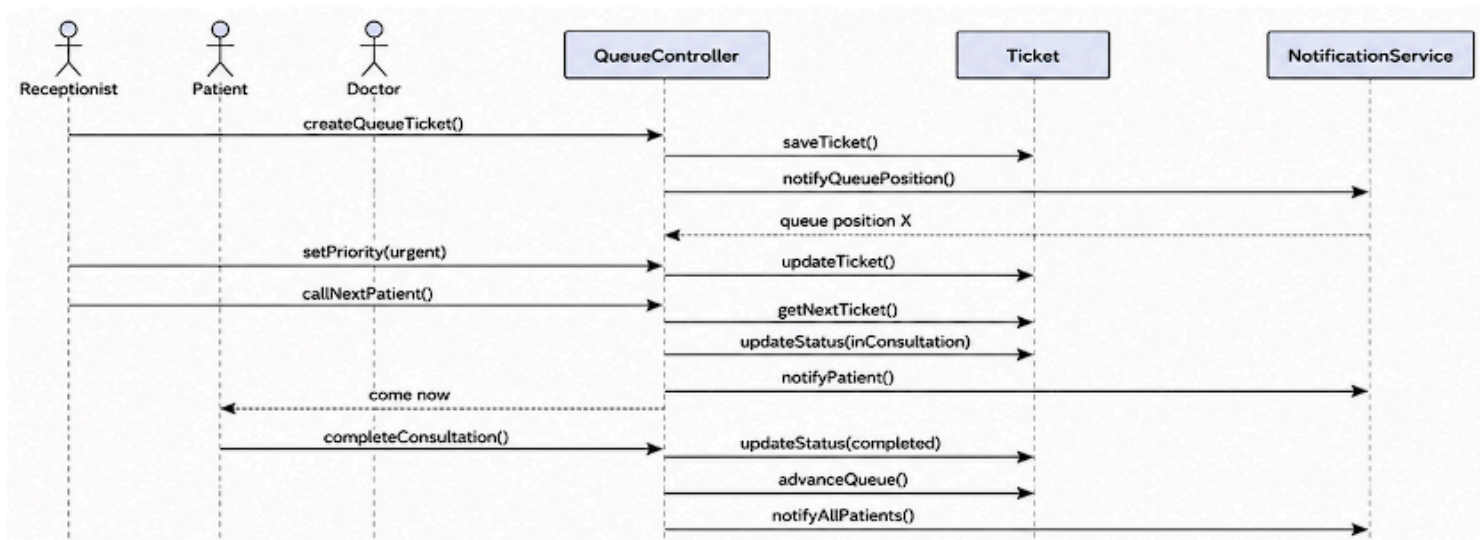
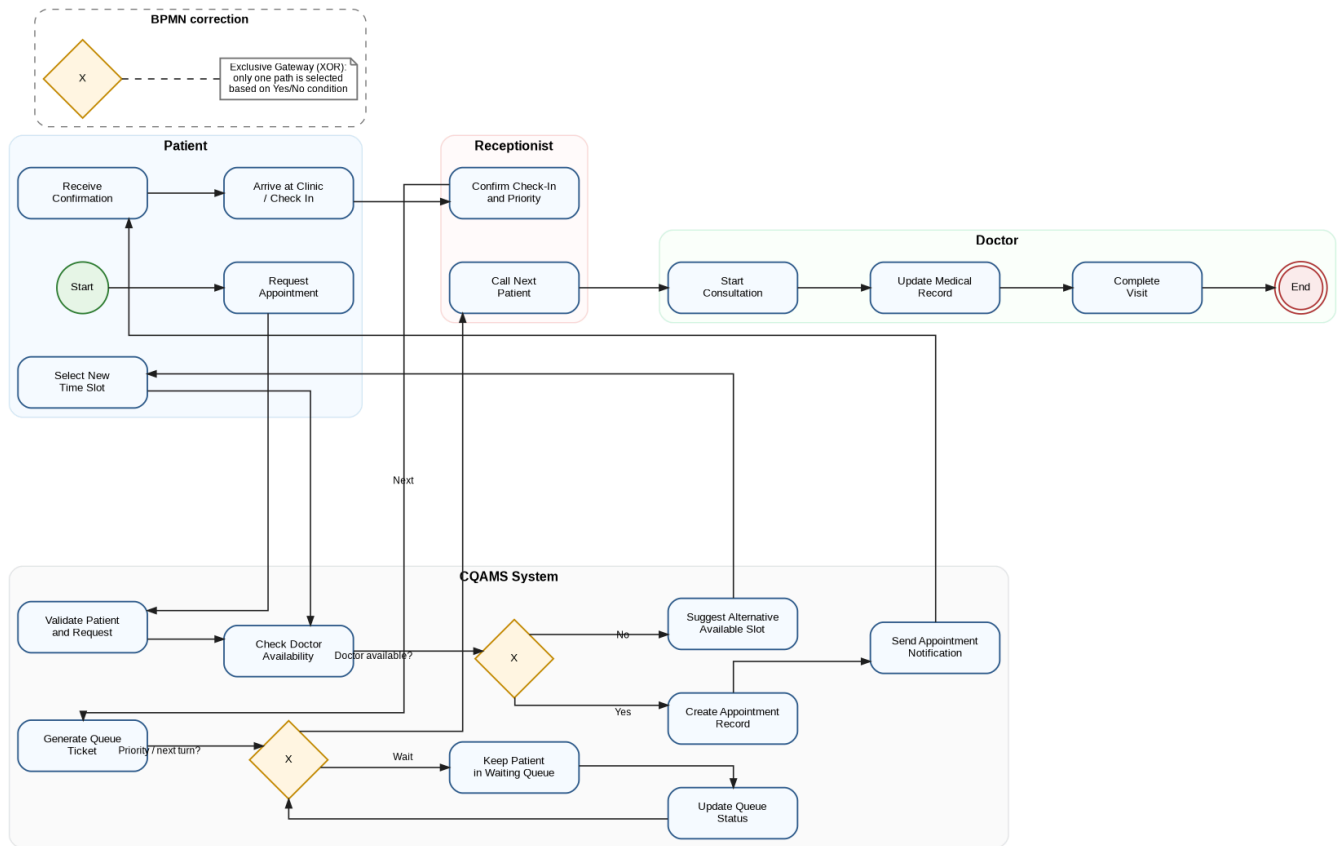


Figure 5.6 - SC-12 Queue Management Sequence Diagram



1. Authentication and Access Control (UC01 - UC06)

Figure 5.6.1 - Manage User Accounts Sequence Diagram

CQAMS Requirements Specification

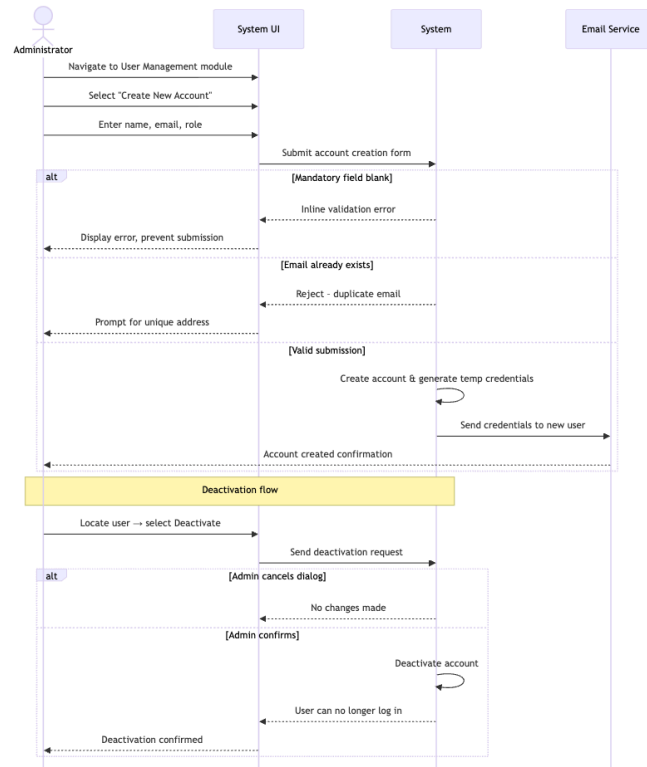


Figure 5.6.2 - Configure Authentication Policies Sequence Diagram

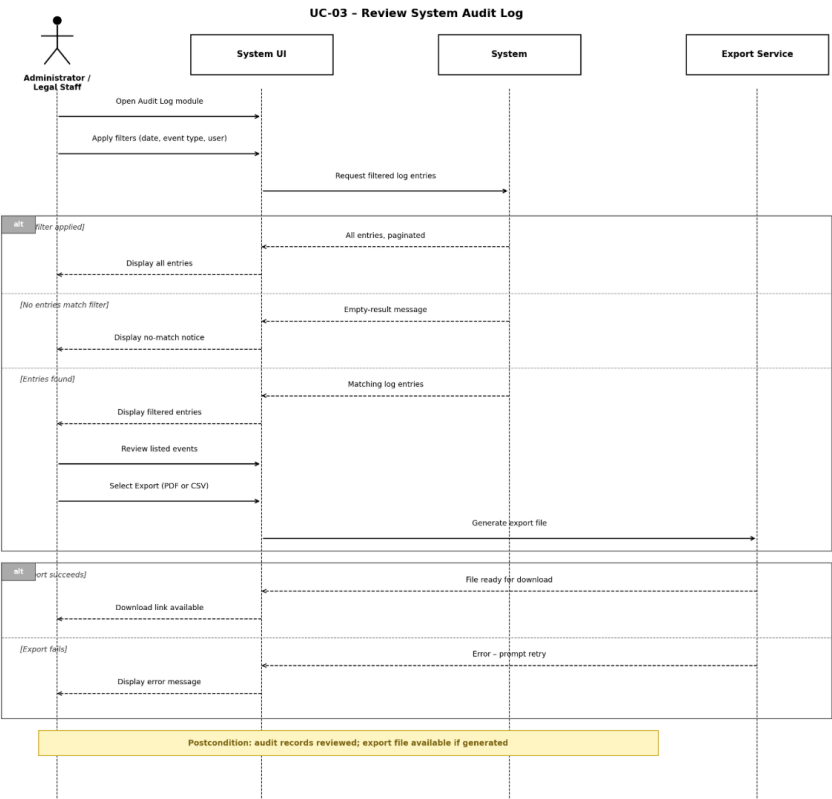


Figure 5.6.3 - Review System Audit Log Sequence Diagram

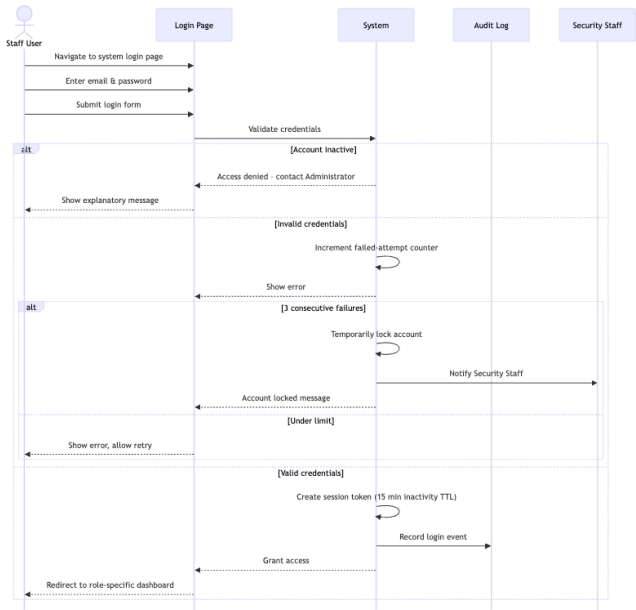


Figure 5.6.4 - User Login Sequence Diagram

CQAMS Requirements Specification

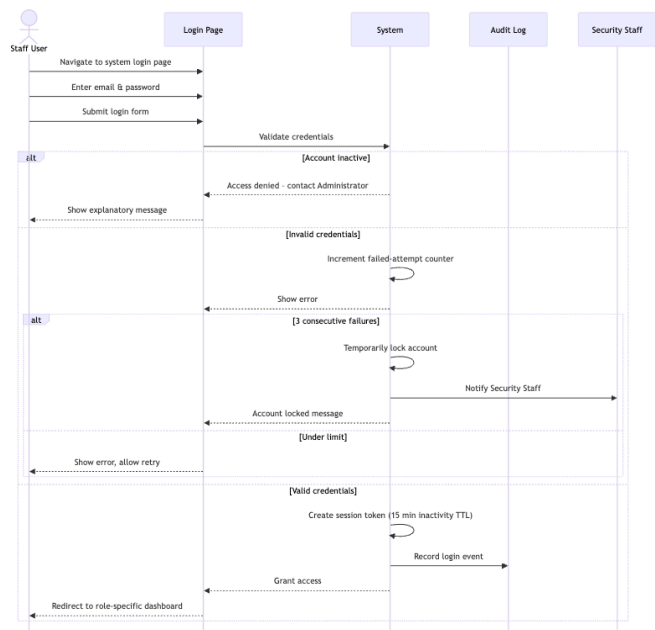


Figure 5.6.5 - Patient Registration and Login Sequence Diagram

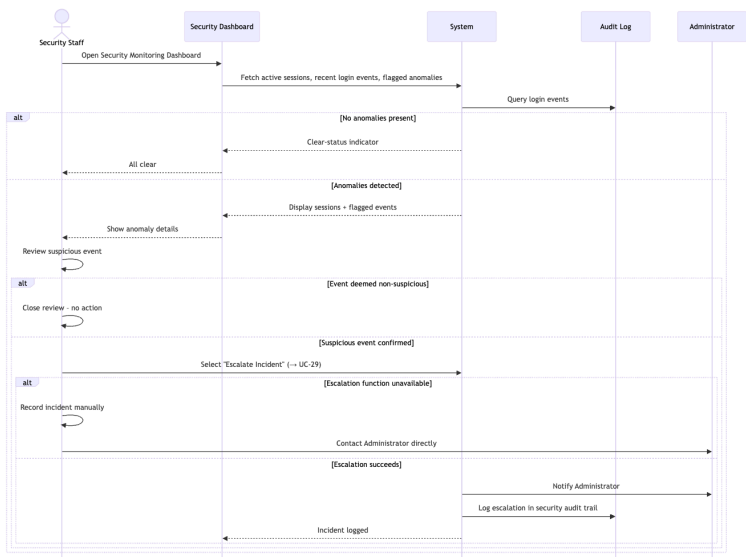


Figure 5.6.6 - Security Staff Monitors Login Events Sequence Diagram

2. Appointment Scheduling (UC07 - UC11)

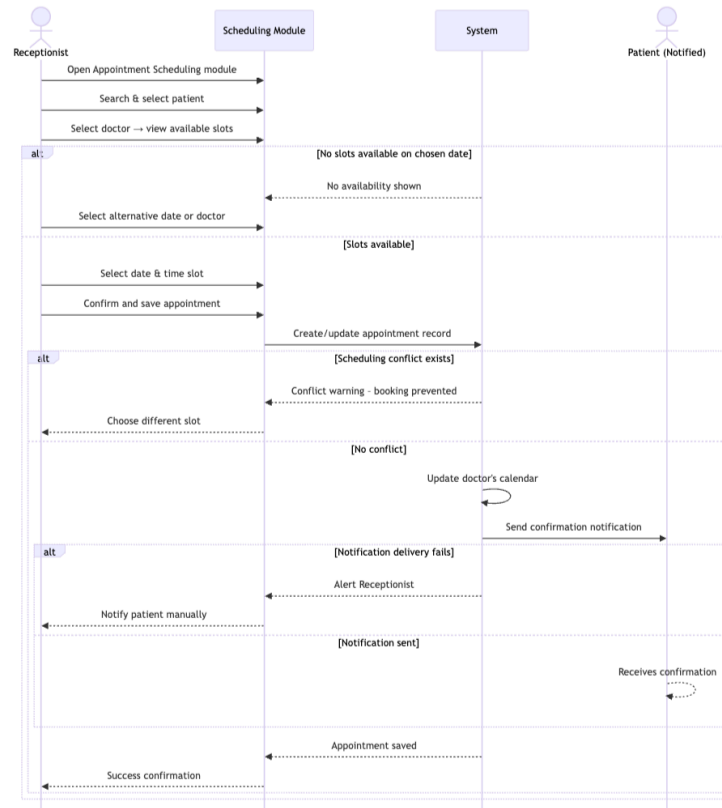


Figure 5.6.7 - Schedule and Modify Appointment Sequence Diagram

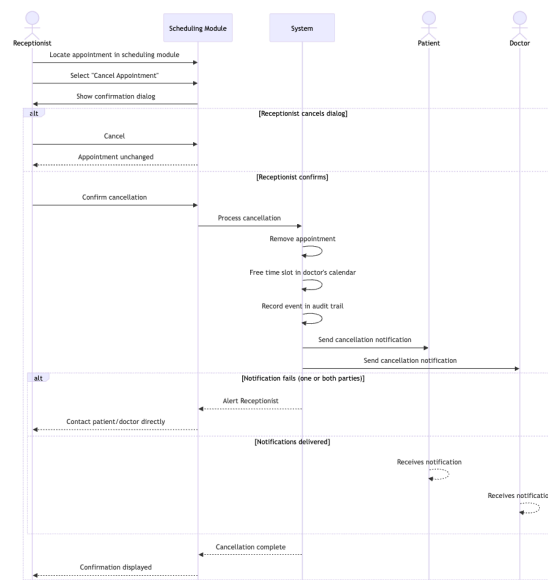


Figure 5.6.8 - Cancel Appointment Sequence Diagram

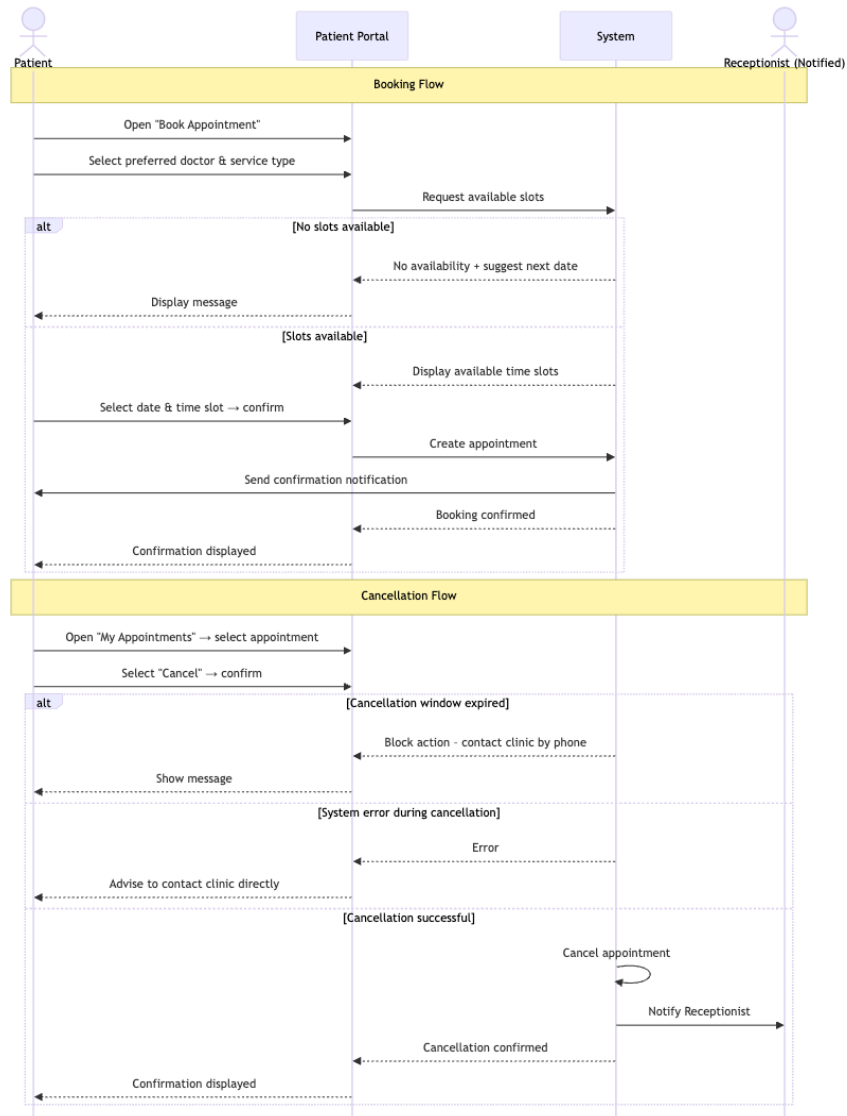


Figure 5.6.9 - Patient Books or Cancels Appointment Sequence Diagram

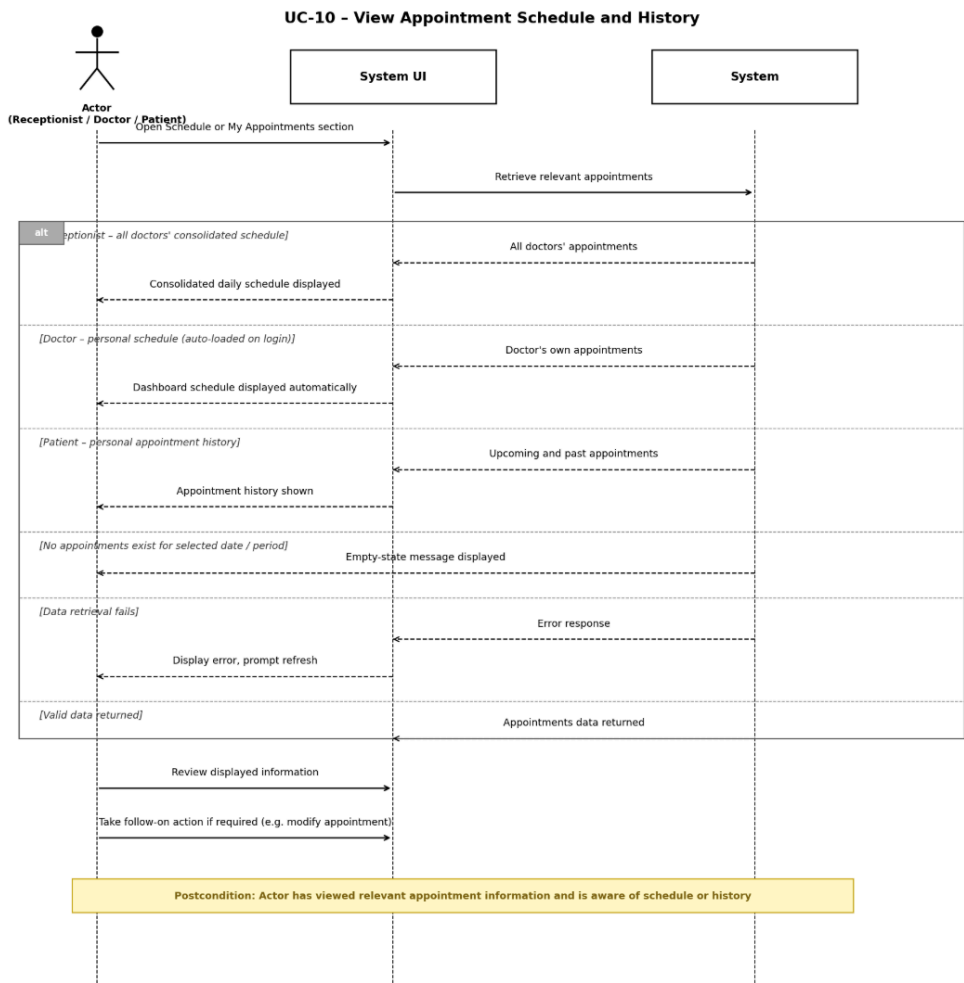


Figure 5.6.10 - View Appointment Schedule and History Sequence Diagram

CQAMS Requirements Specification

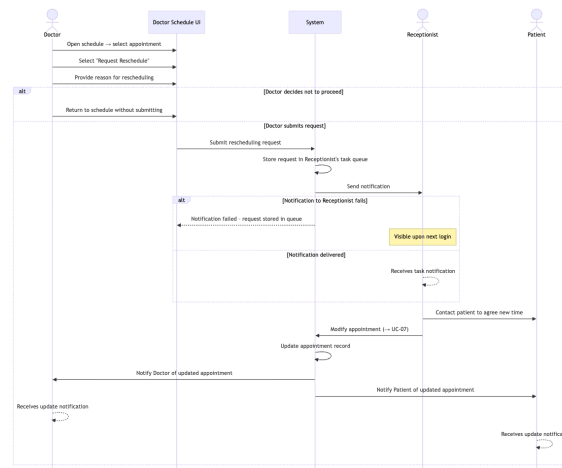


Figure 5.6.11 - Request Appointment Rescheduling Sequence Diagram

3. Queue Management (UC12 - UC16)

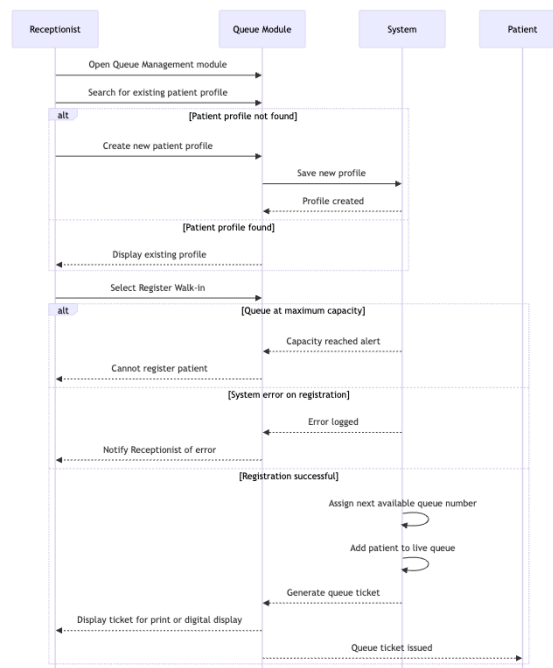


Figure 5.6.12 - Register Walk-in Patient to Queue Sequence Diagram

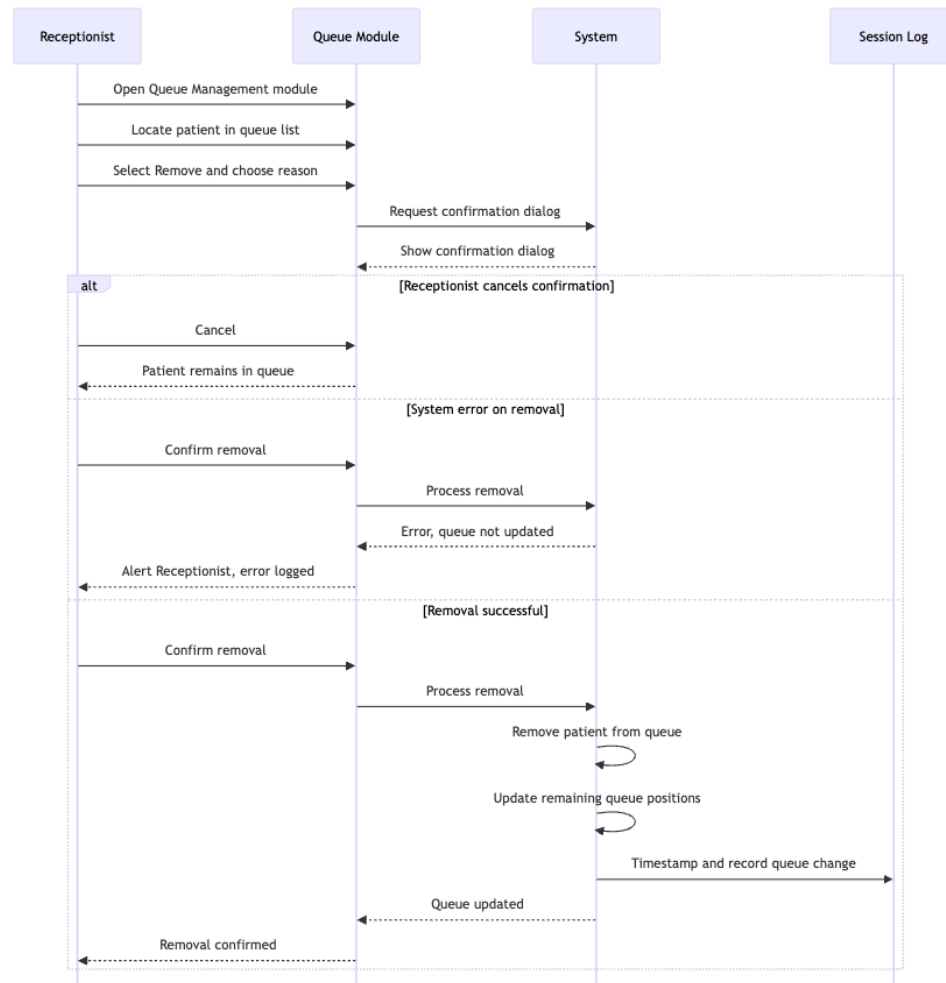


Figure 5.6.13 - Remove Patient from Queue Sequence Diagram

CQAMS Requirements Specification

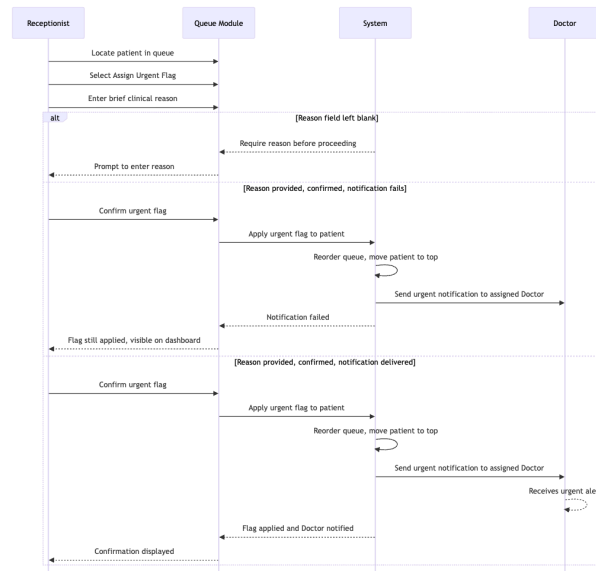


Figure 5.6.14 - Assign Urgent Priority Flag Sequence Diagram

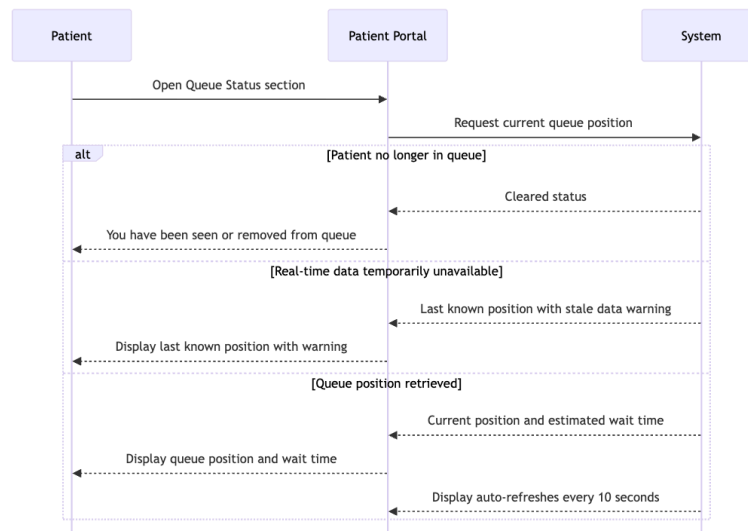
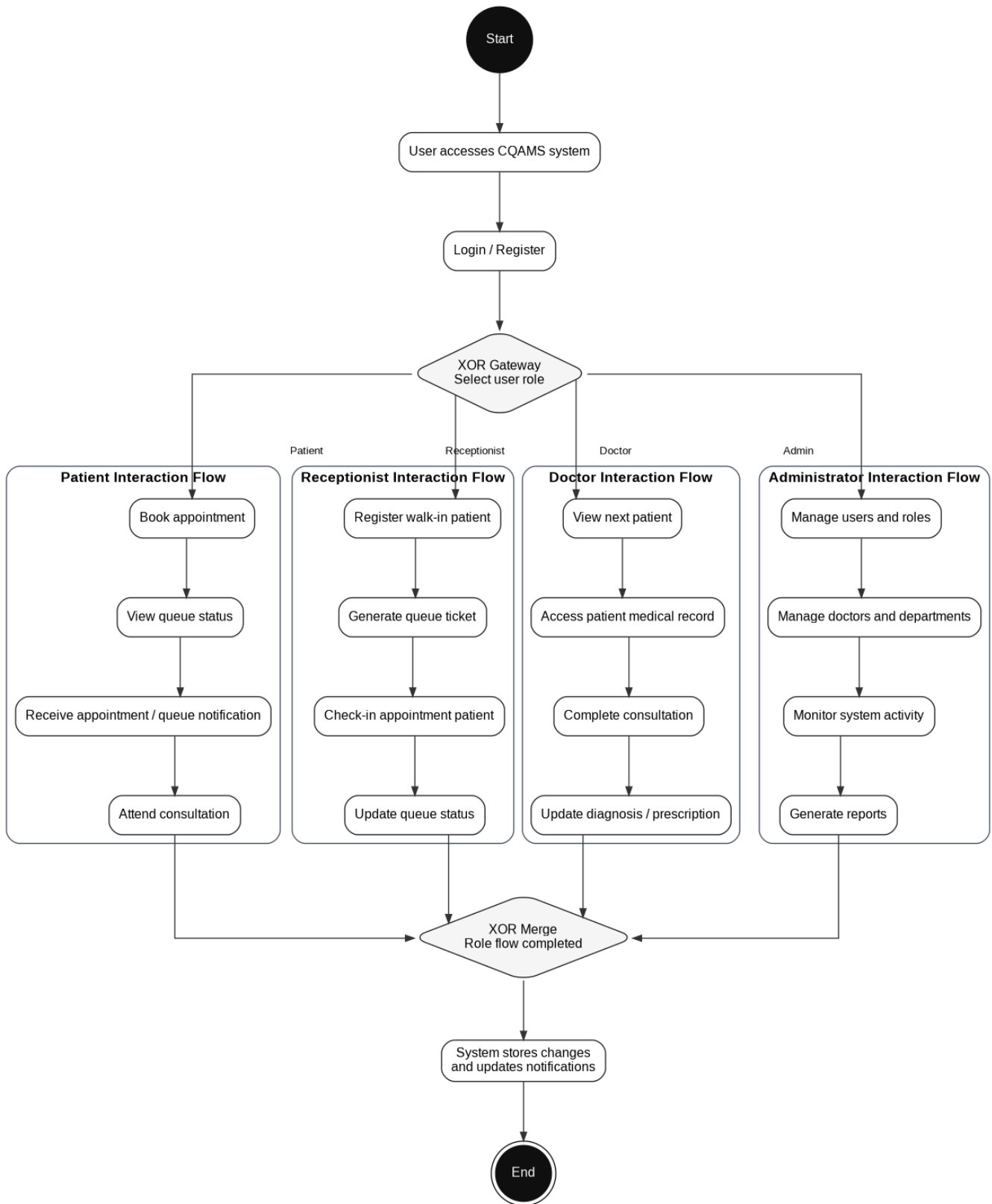
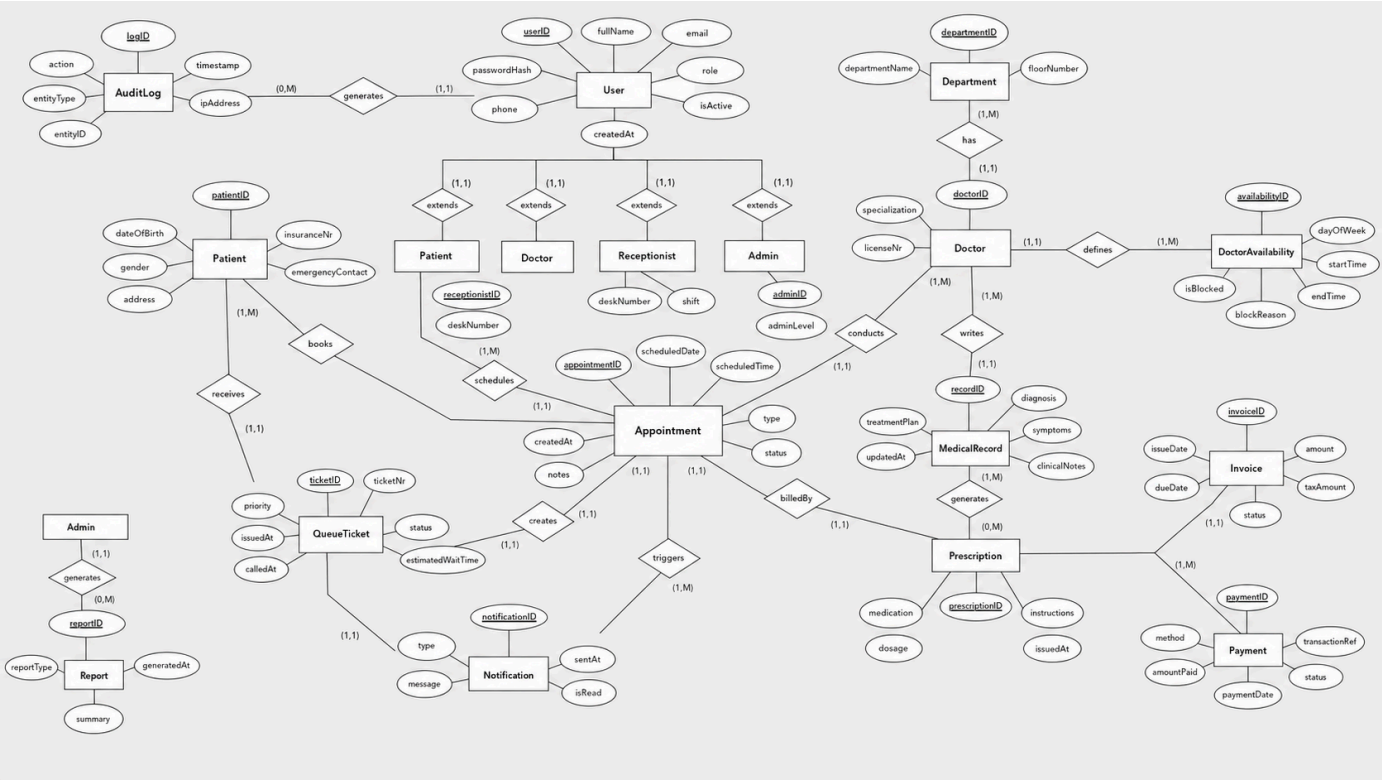


Figure 5.6.15 - Patient Views Queue Position Sequence Diagram



CQAMS Requirements Specification

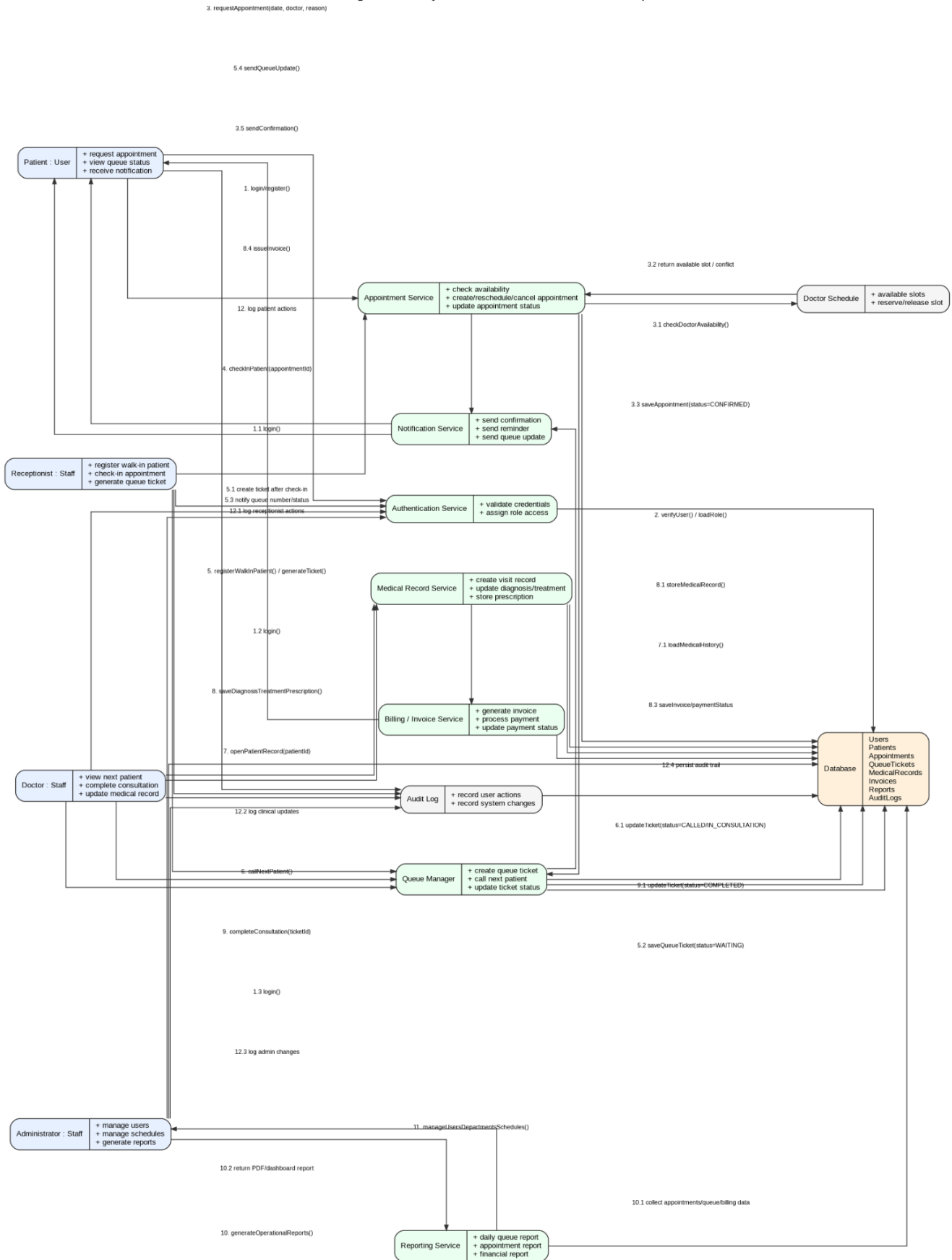


5.7 Collaboration diagram

The collaboration/communication diagram represents the same queue management behaviour as the sequence diagram but focuses on structural object communication and numbered message exchange.

CQAMS Requirements Specification

CQAMS - Detailed Collaboration / Communication Diagram Numbered messages show object interactions across all main requirements



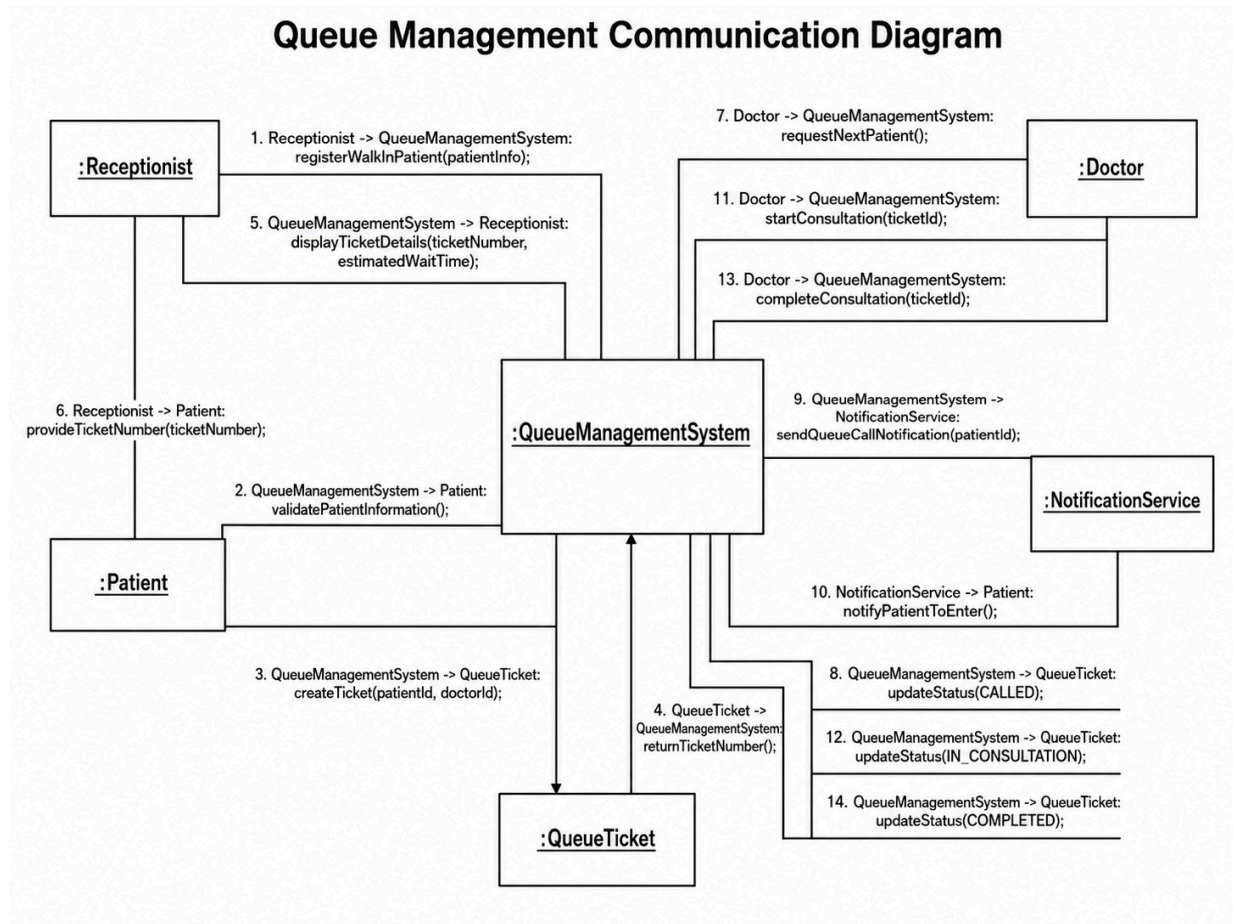


Figure 5.7 - CC-12 Queue Management Collaboration / Communication Diagram

6. Design Patterns

The following design patterns were selected because they directly support CQAMS modules and improve extensibility, maintainability, and separation of concerns. Each pattern includes reasoning and an associated class diagram from the project work.

Pattern	CQAMS Area	Reasoning	Main Classes
Observer Pattern	Used for automatic queue and appointment notifications.	The Observer Pattern is suitable for CQAMS because the system must react automatically when important statuses change. For example, when a queue ticket moves from WAITING to CALLED, the patient must be notified, the rece...	Subject, Observer, QueueTicket, Appointment, PatientNotificationObserver, DoctorNotificationObserver, ReceptionistDashboardObserver, NotificationService.
Factory Method Pattern	Used for role-based user creation.	The Factory Method Pattern is appropriate because CQAMS manages different user roles: Patient, Doctor, Receptionist, and Administrator. These users share common attributes through the abstract User class, but each role h...	User, Patient, Doctor, Receptionist, Administrator, UserFactory, PatientFactory, DoctorFactory, ReceptionistFactory, AdministratorFactory, RegistrationService, ...
Strategy Pattern	Used for flexible queue prioritization rules.	The Strategy Pattern fits CQAMS because the clinic may not always use the same queue ordering rule. In normal cases, patients can be handled using a first-come-first-served strategy. However, emergency cases may need hig...	QueuePriorityStrategy, FirstComeFirstServedStrategy, EmergencyPriorityStrategy, AppointmentTimeStrategy, QueueManagementSystem, QueueTicket, Patient, Doctor.
State Pattern	Used for controlling the queue ticket lifecycle.	The State Pattern is useful because QueueTicket objects move through a defined lifecycle. A ticket may be Waiting, Called, In Consultation, Completed, or Cancelled. Each state allows only certain actions. For example, a ...	TicketState, WaitingState, CalledState, InConsultationState, CompletedState, CancelledState, QueueTicket, QueueManagementSystem.
Facade Pattern (Bonus)	Used to simplify complex clinic workflows.	The Facade Pattern is a strong bonus pattern for CQAMS because many user actions require several internal services. For example, booking an appointment	ClinicWorkflowFacade, ReceptionistUI, PatientPortal, DoctorDashboard, PatientService, AppointmentService,

		may involve checking availability, creating an appointment, updating...	QueueService, MedicalRecordService, NotificationService...
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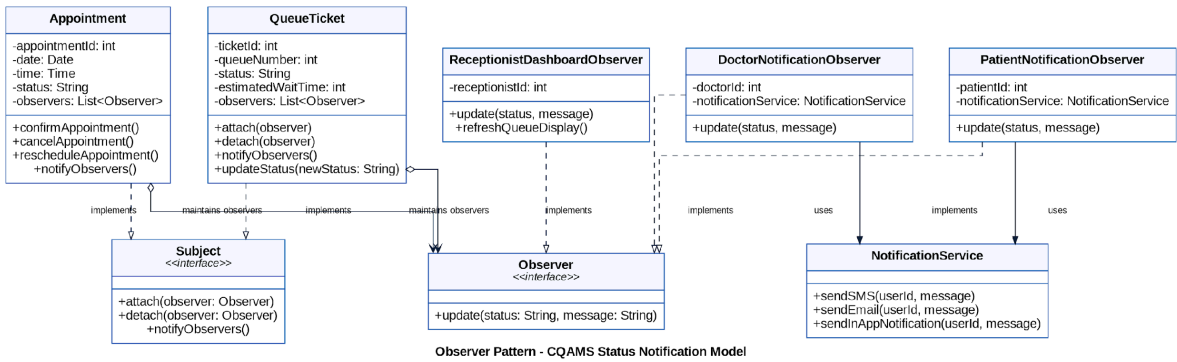


Figure 6.1 - Observer Pattern Class Diagram

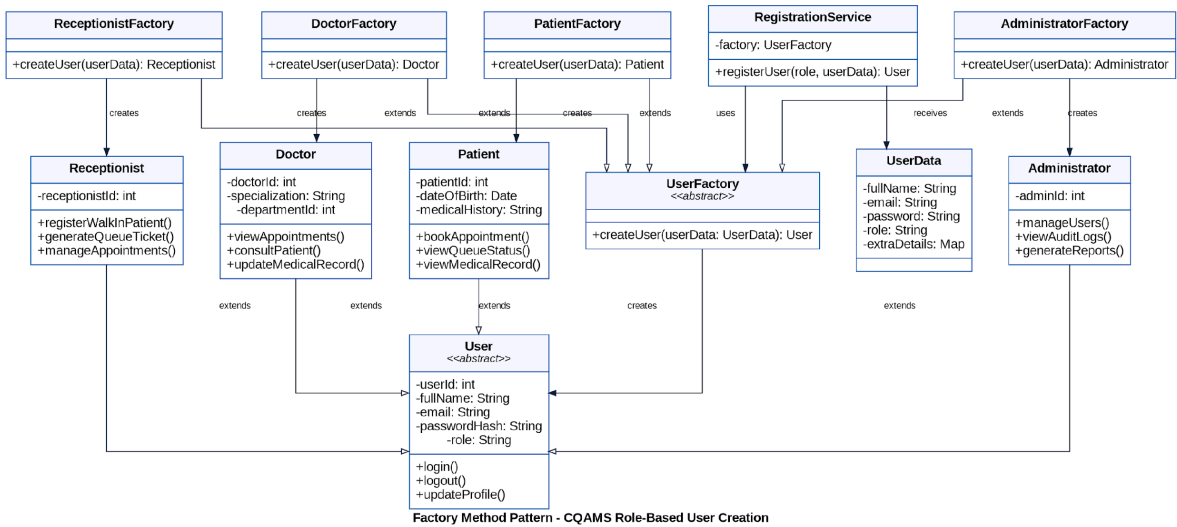
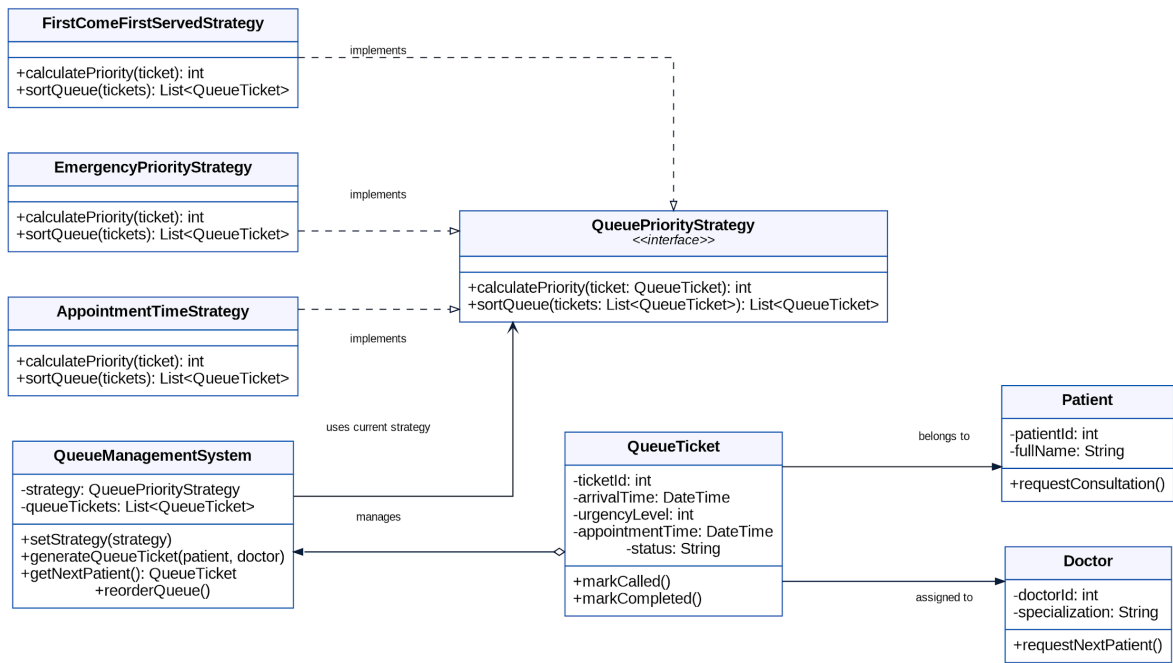


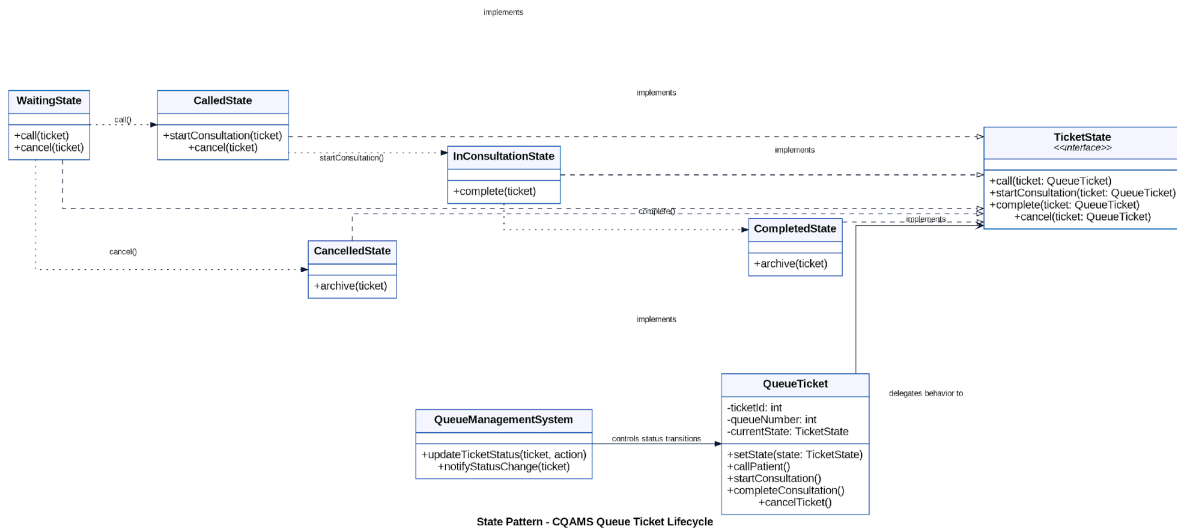
Figure 6.2 - Factory Method Pattern Class Diagram

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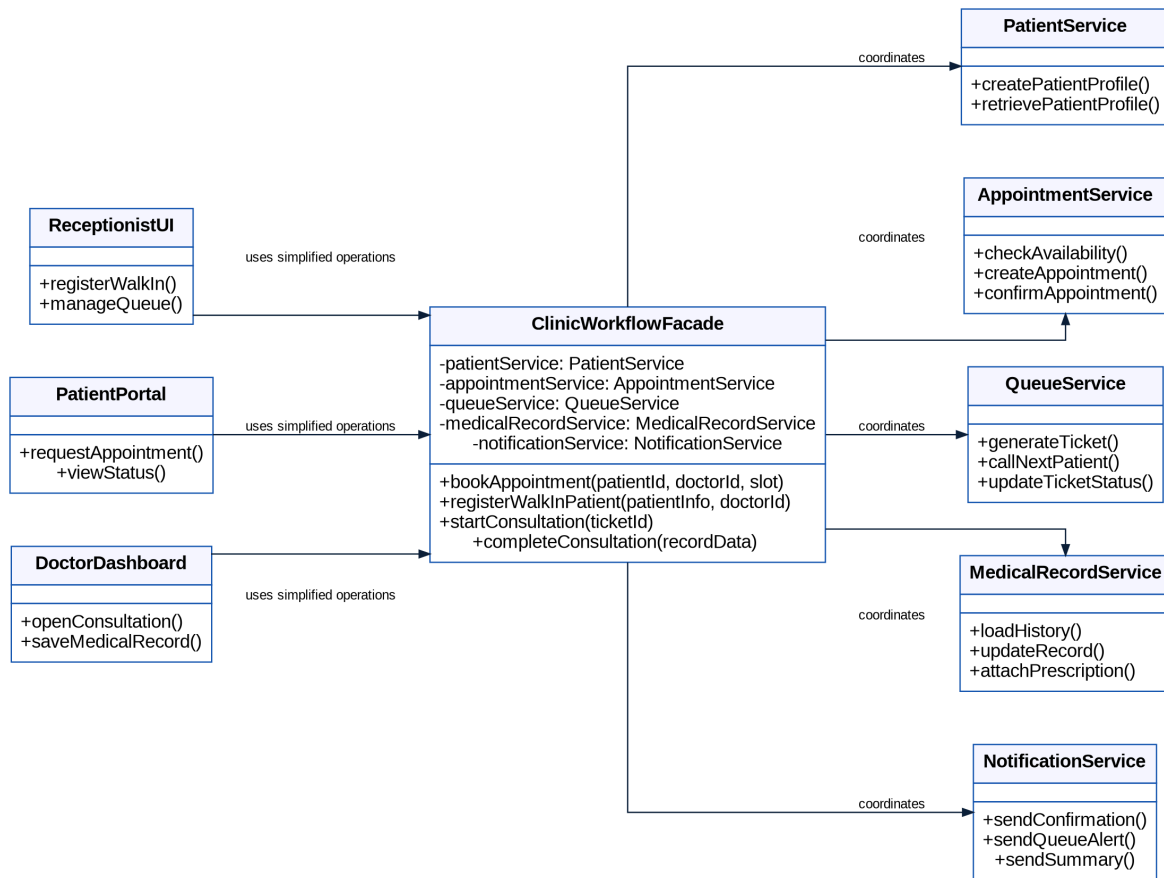
Strategy Pattern - CQAMS Queue Prioritization Rules

Figure 6.3 - Strategy Pattern Class Diagram



State Pattern - CQAMS Queue Ticket Lifecycle

Figure 6.4 - State Pattern Class Diagram



Facade Pattern - CQAMS Simplified Clinic Workflow API

Figure 6.5 - Facade Pattern (Bonus) Class Diagram

7. Appendix.

Appendix A - Traceability Matrix

Functional Domain	Requirements	Use Cases	Diagrams
Authentication and Access Control	FR-01 to FR-10	UC-01 to UC-06	Use Case Diagram; RBAC class structures; audit/security flows
Appointment Scheduling	FR-11 to FR-19	UC-07 to UC-11	AC-07 Appointment Booking BPMN; use case diagram
Queue Management	FR-20 to FR-26	UC-12 to UC-16	AC-12 Walk-in Queue BPMN; SC-12 Sequence; CC-12 Collaboration
Doctor Schedule and Availability	FR-27 to FR-31	UC-17 to UC-19	Availability module in BPMN; Doctor state diagram
Patient Records Management	FR-32 to FR-37	UC-20 to UC-22	Consult Medical Record BPMN; ERD; Medical Record entities
HR and Finance	FR-38 to FR-43	UC-23 to UC-24	Use Case Diagram; organizational workflows
Notifications	FR-44 to FR-49	UC-25 to UC-26	Observer pattern; notification service interactions
Reporting and Administration	FR-50 to FR-55	UC-27 to UC-29	Admin state diagram; audit/security monitoring flows

Appendix B - Source Artefacts Consolidated

- CQAMS_Requirements_Document_v4.docx
- CQAMS_UseCases_v2.docx
- CQAMS_UseCaseDiagram_v2.docx
- CQAMS_Week_StateDiagrams_ERD.docx
- BPMN + Collaboration Diagrams folder
- CQAMS_5_Design_Patterns_Class_Diagrams_FINAL.docx

